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EF(X) Orientations: A Parameterized Complexity Perspective

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Abstract

The concept of fair orientations in graphs was introduced by Christodoulou, Fiat, Koutsoupias and Sgouritsa [23], naturally modeling fair division scenarios in which resources are only contested by neighbors. In this model, vertices represent agents and undirected edges represent goods; edges have to be oriented towards one of their endpoints, i.e., allocated to one of their adjacent agents. Although EFX orientations (*envy-free up to any good*) have been extensively studied in this setting, EF orientations (*envy-free*) remain unexplored. In this work, we initiate their study primarily under the lens of parameterized complexity, presenting various tractable cases, hardness results, and parameterizations. Our results concern both simple graphs and multigraphs. Interestingly, many of our results transfer to EFX orientations, thus complementing and improving upon previous work; notably, we settle an open question regarding the structural parameterized complexity of the latter problem on graphs of polynomially-bounded valuations (Deligkas, Eiben, Goldsmith and Korchemna [27]). We also show that EF orientations are tractable in cases in which EFX orientations are not, particularly for binary valuations. We achieve this through a complete characterization for EF orientations with binary valuations, roughly stating that an EF orientation exists if and only if every connected component of the graph satisfies one among four graph properties. Lastly, we consider *charity* in the orientation setting, establishing algorithms for finding the minimum amount of edges that have to be removed from a graph in order for EF(X) orientations to exist.

1 Introduction

The fair division of indivisible goods is a fundamental problem in computer science, with numerous applications, such as distribution of inheritance, scheduling jobs, and allocating computational resources to machine learning models. The concept of fair division of *divisible* resources was introduced by Steihaus [55]; in recent years, interest has shifted towards *indivisible* goods, due both to the aforementioned applications and to several intriguing open questions surrounding the study of *envy*. Since envy-free (EF) allocations are not guaranteed to exist, research has focused on *envy minimization* (e.g., [42, 47, 52]), and *relaxations* of EF, such as EF1 [18] and EFX [20, 34]. The existence of EFX allocations remains a major open question in fair division (cf. [4]), with some results for two [53] and three agents [3, 21], and further work limiting agents' valuations [5, 6, 35, 38].

A restricted version of the problem was introduced by Christodoulou, Fiat, Koutsoupias and Sgouritsa [23], with a graph model where the vertices represent agents and the edges represent goods; it was proven in the same paper that EFX allocations always exist in this restricted version for simple graphs. This model has been extensively studied since, with the existence of EFX allocations also being studied in multigraphs [1, 2, 9, 43, 54]. An interesting concept in this model is that of *EFX orientations*, i.e., EFX allocations where edges have to be allocated to one of their endpoints, naturally modeling practical applications where resources are contested only by neighbors. EFX orientations have also received significant attention in recent years, with their parameterized complexity in particular being studied in [2, 10, 27].

In spite of this, EF orientations remain largely unexplored. In this work we study them primarily under the lens of parameterized complexity, and present a plethora of tractable special cases, parameterized algorithms, and hardness results, including a complete characterization for the case of binary valuations (Theorem 5). Many of our results transfer to EFX orientations, thus complementing and improving upon results from previous work on EFX orientations [2, 10, 27]; see Section 1.2 for details.

Another interesting relaxation of EF(X) that we consider in this work concerns allocations in which not every item is allocated to an agent, i.e., EF(X) with *charity* [19]. To the best of our knowledge, we are the first to consider charity in the graph setting of [23], establishing algorithms for finding the minimum number of edges that have to remain unoriented (i.e., donated to charity) in order for EF(X) orientations to exist.

1.1 Related Work

Allocations. Regarding EF allocations, NP-completeness and inapproximability results are shown by Lipton, Markakis, Mossel and Saberi [47]. Recent results for the existence of EFX allocations include the following: Plaut and Roughgarden [53] show existence for two agents with general monotone valuations; Chaudhury, Garg and Mehlhorn [21] for three agents with additive valuations; Akrami et al. [3] improve this to a slightly more general class; Prakash, Ghosal, Nimbhorkar and Varma [38] show existence when agents have one of three additive valuations.

Orientations. The existence of EFX orientations is known to be NP-complete even for symmetric valuations by Christodoulou et al. [23]. Deligkas, Eiben, Goldsmith and Korchemna [27] prove that the problem remains NP-complete even for graphs with vertex cover size 8 or multigraphs with 10 vertices, while also identifying one of the few known fixed-parameter tractable cases through the parameterization by *slim tree-cut width* (defined in [33]). Blazej, Gupta, Ramanujan and Strulo [10] study the complexity of the problem in relation to how close the graph is to being bipartite, proving

that it remains NP-complete for (a subclass of) binary instances, while also showing a linear-time algorithm for symmetric binary instances of constant treewidth. Afshinmehr, Danaei, Kazemi, Mehlhorn and Rathi [2] and Hsu [36] further study EFX orientations in (bipartite) multigraphs. Zeng and Mehta [56] relate the existence of EFX orientations to the graph's chromatic number. EFX orientations have also been studied in the setting of chores, i.e., negatively valued items, and mixed manna, i.e., goods and chores [37, 57]. Despite all the aforementioned work for EFX orientations, to the best of our knowledge, the only work relevant to EF orientations appears in [51], where the authors study EF *allocations* in simple graphs, with a polynomial-time algorithm for binary valuations also transferring to *orientations*.

Charity. The concept of charity was first introduced by Caragiannis, Gravin, and Huang [19], in the setting of EFX allocations. Chaudhury, Kavitha, Mehlhorn and Sgouritsa [22] proved that the number of unallocated items that guarantees existence of EFX for general monotone valuations is upper bounded by $n-1$, where n is the number of agents. This bound was later reduced to $n-2$ for additive valuations [7], and extended to general monotone valuations [50]. For 4 agents, this number is 1 [7]. To the best of our knowledge, charity has never been studied in the context of fair orientations prior to this work, although one could obtain a simple charity bound of $|E|/2$ for EFX orientations in simple graphs by removing edges to obtain a bipartite graph [30], which is known to always admit an EFX orientation [56].

1.2 Our Contributions

We start by considering in Section 3.1 the EF Orientation problem[1] in multigraphs with binary valuation functions. We provide a complete characterization of EF orientations in this case (Theorem 5), roughly stating that an EF orientation exists if and only if every connected component of the graph satisfies at least one among four graph properties. Our characterization implies a linear-time algorithm for EF Orientation in the case of binary valuations; in fact, the algorithm can be extended to the more general Min-Charity EF Orientation problem (i.e., finding the minimum amount of edges that have to be removed for an EF orientation to exist). This result contrasts the NP-hardness of EFX Orientation in the analogous setting, which persists even for symmetric instances on simple graphs that are two edge-deletions away from being bipartite [10]. It also generalizes to multigraphs the algorithm of Misra and Sethia [51] for EF Orientation in simple graphs.

In Section 3.2 we present various hardness results for EF Orientation. First we show that the problem remains NP-hard even on 3-regular simple graphs with weights in $\{1, 2\}$ and symmetric valuations, through a reduction from an appropriate variant of SAT. Given this hardness for very restrictive valuations, we next ask whether restricting the *structure* of the input graph allows for more efficient algorithms. Via a simple reduction from Partition we show that, for arbitrary weights, EF Orientation remains NP-hard even for simple graphs of vertex cover number 2 and multigraphs with 2 vertices. Therefore, we move on to consider instances with polynomially-bounded weights, in which case we prove W[1]-, XNLP-, and XALP-hardness for the parameterization by vertex cover, pathwidth, and treewidth respectively (Theorem 12) through a reduction from the Target Outdegree Orientation problem [12, 16]. We thus argue that the problem remains highly intractable even in structured inputs. As a byproduct of our reduction in Theorem 12 and in conjunction with a recent result on the fine-grained complexity of Unary Bin Packing [17], we

[1]For formal definitions of all the problems in question we refer to Section 2.

obtain a $n^{o(\text{vc})}$ ETH-based lower bound, where vc denotes the vertex cover number of the input graph, that holds even for simple graphs.

Although the hardness results of Section 3.2 seem very discouraging, we next identify restricted cases on which we are able to obtain fast algorithms. First, we consider instances on simple graphs with at most k *heavy* edges, that is, edges that have valuation at least 2 for an endpoint; this naturally generalizes the binary-valuation regime considered in Section 3.1, albeit restricted to simple graphs. Here we develop a $2^k n^{\mathcal{O}(1)}$ algorithm, which we complement with a tight SETH-based lower bound (Theorems 9 and 13). We view this as a first step towards generalizing the algorithm of Theorem 6 to an FPT algorithm parameterized by the number of heavy edges. Second, using dynamic programming (DP), we show that EF Orientation can be solved in time $(n+m)^{\mathcal{O}(\text{tw})}$ for polynomially-bounded weights, where tw denotes the treewidth of the graph and m the number of edges. Our algorithm works for the more general Min-Charity EF Orientation problem in multigraphs with non-symmetric valuations, and matches the lower bound of Theorem 12.

EF Orientation is arguably a very natural, understudied problem and our results, apart from being interesting in their own right, additionally pave the way to advance the state-of-the-art for the closely related, and better understood, EFX Orientation problem. In particular, in Section 4.1 we present a reduction from EF Orientation to EFX Orientation that allows us to transfer most hardness results of Section 3.2 to the latter. Consequently, we obtain NP-hardness for simple graphs of vertex cover number 4 and multigraphs of 4 vertices, improving upon the hardness results of Deligkas et al. [27] for constants 8 and 10 respectively, as well as a hardness result by Afshinmehr et al. [2] for multigraphs with 8 vertices. More importantly, we obtain a $n^{o(\text{vc})}$ ETH-based lower bound for simple symmetric instances with polynomially-bounded weights. As a matter of fact, in Section 4.2 we yet again employ DP and obtain an $(n+m)^{\mathcal{O}(\text{tw})}$ algorithm, this time for the Min-Charity EFX Orientation problem; once again, our algorithm works for multigraphs and non-symmetric valuations. Since this matches the aforementioned lower bound, we settle the open question of Deligkas et al. [27] on the complexity of EFX Orientation for instances with polynomially-bounded weights and small vertex cover, and paint a complete picture on the parameterized complexity of EFX Orientation parameterized by treewidth, arguably the most well-studied structural parameter.

We refer to Tables 1 and 2 for a summary of our algorithms and hardness results respectively.

	Running time	**Conditions**
EF	$\mathcal{O}(n+m)$	Binary valuations
EF	$2^k n^{\mathcal{O}(1)}$	At most k heavy edges Simple graph
EF(X)	$W^{\mathcal{O}(\text{tw})}(n+m)^{\mathcal{O}(1)}$	W: *max. shared weight* tw: treewidth
EF(X)	$(n+m)^{\mathcal{O}(\text{tw})}$	Poly weights tw: treewidth

Table 1: Our algorithms for EF and EFX Orientation in (multi)graphs with n vertices and m edges (Sections 3.1, 3.3, 3.4, and 4.2). All algorithms except the one in the second row also extend to the respective minimum-charity problem.

	Hardness/ Lower Bound	Conditions
EF	no $(2-\varepsilon)^k n^{\mathcal{O}(1)}$ time algorithm (under SETH)	At most k heavy edges Polynomial weights
EF	NP-complete	3-regular graphs Weights in {1,2}
EF	NP-complete (weakly)	Vertex cover 2 **or** Multigraph with 2 vertices
EFX	NP-complete (weakly)	Vertex cover 4 **or** Multigraph with 4 vertices
EF(X)	W[1]-hard	Parameter: vertex cover number
	XNLP-hard	Parameter: pathwidth
	XALP-hard	Parameter: treewidth
		All three: Polynomial weights
EF(X)	no $n^{o(\mathrm{vc})}$ time algorithm (under ETH)	vc: vertex cover number Polynomial weights

Table 2: Our hardness results for EF and EFX Orientation (Sections 3.2 and 4.1 respectively). All of these results hold even for **simple symmetric** instances, except the "multigraph" results in the third/fourth rows, which are restricted just to symmetric instances.

2 Preliminaries

Throughout the paper we use standard graph notation [28] and we assume familiarity with the basic notions of parameterized complexity (cf. Section A). For a graph G and $S \subseteq V(G)$ a subset of its vertices, $G[S]$ denotes the subgraph induced by S while $G - S$ denotes $G[V \setminus S]$. For $x, y \in \mathbb{Z}$, let $[x, y] = \{z \in \mathbb{Z} \mid x \leq z \leq y\}$ while $[x] = [1, x]$. Proofs of statements marked with (★) are deferred to the appendix.

Graphs in the context of this paper are *multigraphs*, i.e., parallel edges between two vertices may exist, unless we explicitly state that a graph is *simple*. All graphs considered are undirected and without loops. We denote the set of edges between vertices u, v of a multigraph as E_{uv}, and the set of edges incident to u as E_u.

2.1 Model and Problem Definitions

We are given a set of agents $V = \{1, \ldots, n\}$ and a set of items E with $|E| = m$. Each agent $i \in [n]$ has a valuation function $v_i \colon 2^E \to \mathbb{N}$. An *allocation* is a partition of E into bundles $X_1, \ldots, X_n$, where agent i receives the bundle X_i. A *partial allocation* is an allocation of some subset of E.

In the context of this work, we only consider *additive* valuation functions, as is standard for most of the fair orientation literature: each v_i is defined over E, and can be extended to 2^E with $v_i(X) := \sum_{e \in X} v_i(e)$, $\forall X \subseteq E$. As is standard, we assume $v_i(\varnothing) = 0$ for all $i \in V$.

Definition 1 (Envy)**.** *Given an allocation $X_1, \ldots, X_n$, we say that agent i* envies *j if $v_i(X_i) < v_i(X_j)$. We say that i* strongly envies *j if $v_i(X_i) < v_i(X_j \setminus e)$ for some $e \in X_j$.*

Definition 2 (Fair allocations)**.** *We say that an allocation is* envy-free (EF) *if no agent envies another agent. We say that an allocation is* envy-free up to any good (EFX) *if no agent strongly envies another agent.*

In this paper, we consider the model introduced by [23]: we are given sets of agents V and items E that form a multigraph $G = (V, E)$, where for all edges $e \in E_{ij}$ $(i, j \in V)$ it holds that $v_u(e) = 0$, $\forall u \in V \setminus \{i, j\}$.[2] Throughout the paper we may call the agents *vertices* and the goods *edges*. We denote $n = |V|$ and $m = |E|$.

An *orientation* is an allocation $X_1, \ldots, X_n$, where for all $e \in E_{ij}$ $(i, j \in V)$ it holds that $e \in X_i$ or $e \in X_j$. A *partial orientation* is an orientation of some subset of E. Given an orientation, we say that $e \in E_{ij}$ $(i, j \in V)$ is oriented towards i if $e \in X_i$, i.e., if i receives e. An EF(X) orientation is an EF(X) allocation that is an orientation.

We now define the problems which we study in this work.

EF Orientation

Instance: A multigraph $G = (V, E)$ and values $v_i(e), v_j(e)$ for each $e \in E_{ij}$ $(i, j \in V)$.
Goal: Determine whether G admits an EF orientation.

Min-Charity EF Orientation

Instance: A multigraph $G = (V, E)$ and values $v_i(e), v_j(e)$ for each $e \in E_{ij}$ $(i, j \in V)$.
Goal: Find the smallest integer $k \geq 0$ s.t. G admits a partial EF orientation of size $|E| - k$.

Intuitively, the Min-Charity EF Orientation problem asks for the minimum amount of edges that must be removed from a graph for an EF orientation to exist. Note that any algorithm for Min-Charity EF Orientation also answers EF Orientation, hence, throughout the paper we may omit an EF Orientation algorithm and directly state the (more general) Min-Charity EF Orientation algorithm.

The EFX Orientation and Min-Charity EFX Orientation problems are defined similarly.

We call an instance of any of these problems

- *simple*, if for all $i, j \in V$: $|E_{ij}| \leq 1$,
- *symmetric*, if for all $i, j \in V$ and all $e \in E_{ij}$: $v_i(e) = v_j(e)$. In that case, it suffices to store one valuation function $v \colon E \to \mathbb{N}$, where for all $i, j \in V$ and all $e \in E_{ij}$: $v(e) = v_i(e) = v_j(e)$.

Throughout the paper, we may refer to the valuation functions in the context of graphs as *weights*: each edge is represented by two weights in the general case (one for each of its endpoints), however one weight per edge suffices if the instance is symmetric.

3 EF Orientations

In this section we present our results concerning EF orientations. The following observation will be useful throughout the section.

Observation 3. *Edges with zero value for both endpoints can be removed from the input without affecting the existence of EF orientations. Similarly, edges with zero value for one endpoint and*

[2] For simplicity, we use the notation $v_u(e)$ instead of $v_u(\{e\})$ to denote the valuation of a single edge e for vertex u.

non-zero value for the other can always be oriented towards the latter without affecting the existence of an EF orientation.[3]

3.1 Algorithms for Binary Valuations

In this subsection we restrict the EF Orientation problem to binary valuations, i.e., we assume that $v_i(e),\ v_j(e) \in \{0,1\}$ for all $e \in E_{ij}$, $i,j \in V$. By Observation 3, we remove from the input all edges of weight 0 for both endpoints and orient all edges of weight 0 for one endpoint and weight 1 for the other endpoint towards the latter.

Observation 4 is the key to proving (part of) our main theorem in this subsection (Theorem 5), which is essentially a complete characterization of EF orientations with binary valuations.

Observation 4. *For an* EF Orientation *instance with binary valuations, define $E' \subseteq E$ to be the subset of the edges with weight* 1 *for both their endpoints. Suppose we have a partial EF orientation where some vertex i has received an edge $e^* \in E_{ij}$ s.t. $v_i(e^*) = 1$. Then, all edges $e \in E'_{ik}$ for $k \neq j$ can be oriented as follows while preserving envy-freeness. Let $p = |E'_{ik}|$.*

- *If p is even, orient $p/2$ edges towards i and $p/2$ edges towards k. It is clear that neither i nor k envy each other.*
- *If p is odd, orient $(p-1)/2$ edges towards i and $(p-1)/2+1$ edges towards k. It is clear that k does not envy i; however, it also holds that i does not envy k because i has received $e^* \in E_{ij}$ with $v_i(e^*) = 1$ and $j \neq k$ by assumption.*

Additionally, if $v_j(e^*) = 0$, then the same observation applies even for $k = j$.

Theorem 5 (Characterization ★)**.** *Let $\mathcal{I} = (G = (V,E), \{v_i\}_{i \in V})$ be an instance of* EF Orientation *with binary valuations and define $G' = (V, E')$ as the subgraph of G s.t. $E' \subseteq E$ consists only of the edges in E with weight* 1 *for both their endpoints. $\mathcal{I}$ has an EF orientation if and only if for every connected component C of G' at least one of the following properties holds.*

1. *C contains a circuit*[4] *consisting of at least* 3 *vertices.*
2. *C contains a vertex i incident with some edge $e \in E_{ij}$ s.t. $v_i(e) = 1$ and $v_j(e) = 0$.*
3. *C contains two vertices i, j s.t. $|E'_{ij}|$ is even and non-zero.*
4. *C consists of only one vertex.*

We refer to Figures 1 and 2 for some intuition for the proof of Theorem 5.

Using Theorem 5, we can obtain the following.

Theorem 6 (★)**.** *There is an algorithm running in time $\mathcal{O}(n+m)$ for* EF Orientation *with binary valuations.*

Theorem 6 contrasts the complexity of EFX Orientation, which is known to be NP-hard even for binary valuations by Blazej et al. [10]. Additionally, it generalizes to multigraphs the algorithm of Misra and Sethia [51], which works only for simple graphs. We next extend this linear-time algorithm to the Min-Charity EF Orientation problem.

[3] Note that neither of these properties hold for EFX orientations.

[4] In graph theory, a circuit is defined as a closed trail where a sequence of vertices starts and ends at the same vertex without repeating any edges. Informally, it is a cycle that may pass through each vertex more than once, while only using each edge at most once.

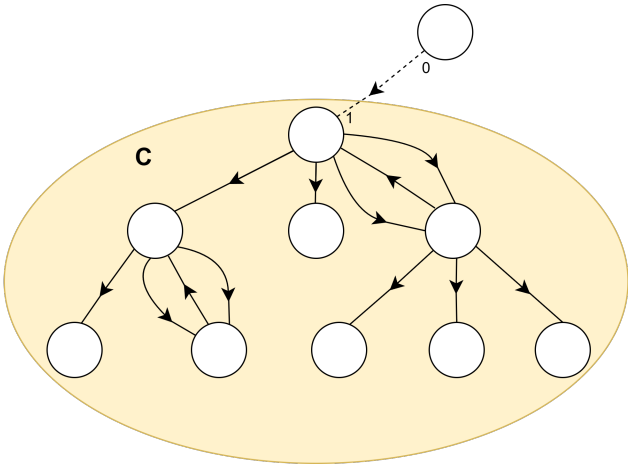

Figure 1: An example of a connected component C of G' that satisfies Property 2 of Theorem 5, i.e., some vertex receives an edge of weight 1 for itself and weight 0 for the other endpoint. This allows us to repeatedly apply Observation 4 starting from that vertex, orienting all edges of C to obtain an EF orientation as demonstrated.

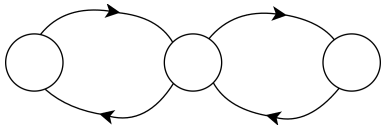

Figure 2: An example of a circuit consisting of 3 vertices, without containing a cycle of length greater than 2. Observe that no vertex is envious of another if the edges are oriented along a cyclic route as shown (assuming all edges have weight 1 for both endpoints).

Theorem 7 (★)**.** *There is an algorithm running in time $\mathcal{O}(n+m)$ for* Min-Charity EF Orientation *with binary valuations.*

Remark 8. The algorithms of Theorems 6 and 7 can also find the respective orientation (if it exists) in time $\mathcal{O}(n+m)$, because the proof of Theorem 5 is constructive, i.e., it orients the whole graph by repeatedly applying Observation 4.

3.2 Hardness Results

Here we present various hardness results concerning the EF Orientation problem. As our first result, we present a lower bound based on the Strong Exponential Time Hypothesis (SETH), which states that SAT does not admit any $(2-\varepsilon)^n n^{\mathcal{O}(1)}$ algorithm, where n is the number of variables. In particular, this shows the optimality (up to polynomial factors) of the algorithm presented in Theorem 13.

Theorem 9 (★)**.** *There is no $(2-\varepsilon)^k n^{\mathcal{O}(1)}$ algorithm for* EF Orientation *under SETH, where k is the number of edges of value at least 2, even when restricted to simple symmetric instances with polynomially-bounded weights.*

We refer to Figure 3 for a sketch of our reduction from SAT in Theorem 9. Next, by adapting essentially the same reduction from a more appropriate variant of SAT, we obtain NP-hardness for EF Orientation even for very restricted valuation functions.

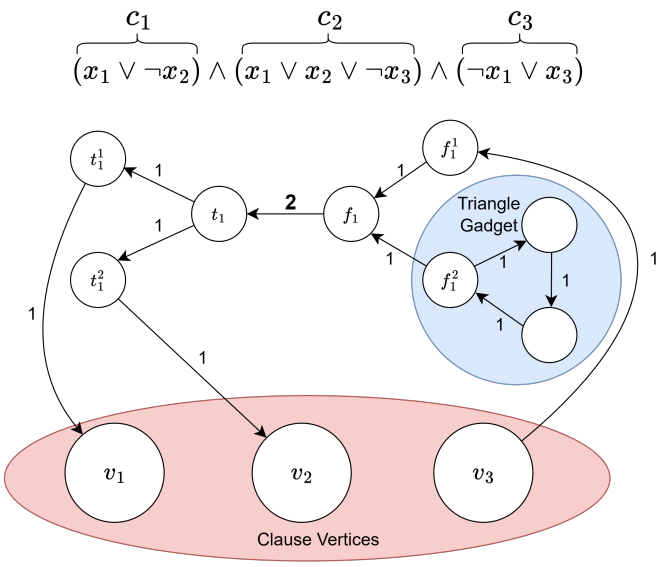

Figure 3: A sketch of our reduction from SAT to EF Orientation in Theorem 9. For simplicity, we only show the part of the graph corresponding to variable x_1. The weighted edge $\{t_1, f_1\}$ is oriented towards t_1 if x_1 is set to true, or towards f_1 if x_1 is set to false. We prove that clause vertex v_i is non-envious of all its neighbors iff clause c_i is satisfied. Observe that v_1 and v_2 are non-envious when $\{t_1, f_1\}$ is oriented towards t_1 (equivalently, c_1, c_2 are satisfied by setting x_1 to true). For more details, see Appendix B.

Theorem 10 (★)**.** EF Orientation *is NP-complete even when restricted to simple symmetric instances on* 3*-regular graphs and weights in* $\{1, 2\}$*.*

Moving on, we show that EF Orientation remains highly intractable even on very restricted classes of (multi)graphs. To show this we first present a simple reduction from Partition, and then one from the Target Outdegree Orientation problem [12, 16].

Theorem 11 (★)**.** EF Orientation *is weakly NP-complete even restricted to*

- *simple symmetric instances of vertex cover number* 2*,*
- *symmetric instances with* 2 *vertices.*

Theorem 12 (★)**.** EF Orientation *is W[1]-hard parameterized by vertex cover number, XNLP-hard parameterized by pathwidth, and XALP-hard parameterized by treewidth, even restricted to simple symmetric instances with polynomially-bounded weights. Furthermore, for such instances,* EF Orientation *does not admit a* $n^{o(\mathrm{vc})}$ *algorithm under the ETH, where* vc *denotes the vertex cover number of the input graph.*

3.3 Parameterization by Number of Heavy Edges

Here we consider EF Orientation on simple graphs. Observe that most of the hardness results of Section 3.2 already apply even under this restriction. Our main result is an algorithm with running time $2^k n^{\mathcal{O}(1)}$, where k denotes the number of edges that have value at least 2 for at least one endpoint; by Theorem 9, this is optimal (up to polynomial factors) under the SETH. We remark that our algorithm works even for binary-encoded weights on the edges.

The rough idea of our algorithm is the following. As a first step we fix an orientation of all "heavy" edges, that is, edges of value at least 2 for some endpoint. This leaves a residual instance in which every unoriented edge has value 1 for both endpoints. To appropriately orient these remaining edges, we reduce to an equivalent instance of Upper Degree-Constrained Graph Orientation, for which a polynomial-time algorithm is known [31]. This allows us, for each fixed orientation of the heavy edges, to decide in polynomial time whether it can be extended to an envy-free orientation. Finally, iterating over all possible orientations of the heavy edges (of which there are at most 2^k) suffices to determine whether the instance admits an envy-free orientation.

Theorem 13 (★). *There is an algorithm for* EF Orientation *on simple graphs running in time* $2^k n^{\mathcal{O}(1)}$*, where* k *is the number of edges with value at least* 2 *for at least one endpoint.*

3.4 Parameterization by Treewidth plus Maximum Shared Weight

At this point, we have shown that EF Orientation remains computationally hard even when restricted to simple graphs of small vertex cover and symmetric valuations: Theorem 11 shows weak NP-hardness for graphs of vertex cover 2, while for polynomially-bounded weights Theorem 12 gives a $n^{o(\mathrm{vc})}$ lower bound under the ETH. A natural question would be to consider whether we can match the latter lower bound with an algorithm. We answer this positively: using standard DP techniques, one can get an algorithm for EF Orientation running in time $W^{\mathcal{O}(\mathrm{tw})} n^{\mathcal{O}(1)}$, where W denotes the maximum weight of the graph and tw its treewidth. For polynomially-bounded weights, this algorithm matches the aforementioned lower bound (see Remark 16).

Pushing this idea further, we observe that in fact similar techniques apply even for multigraphs with non-symmetric valuations, under the caveat that in this setting W is a bound on the sum of the weights of a set of parallel edges for one of its endpoints; observe that this is a straightforward generalization of the maximum weight considered in the previous setting. We define this to be the *maximum shared weight* of an instance. As a matter of fact, our main result in this section is to show that we can solve the (more general) Min-Charity EF Orientation problem in essentially the same running time.

Theorem 14 (★). *Given an instance* $\mathcal{I}$ *of* Min-Charity EF Orientation *along with a nice tree decomposition of the input multigraph of width* tw*, one can decide* $\mathcal{I}$ *in time* $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$*, where* W *denotes the maximum shared weight of* $\mathcal{I}$ *and* n *the number of vertices.*

Proof sketch. Our algorithm is a standard DP over a nice tree decomposition of the input graph. For a node t of the given nice tree decomposition, let G_t denote the corresponding subgraph, while B_t denotes the set of vertices appearing in the bag of node t. For every node t of the tree decomposition, we keep track of whether there exists a partial orientation of G_t (excluding the edges among vertices of B_t) that guarantees that the vertices of $V(G_t) \setminus B_t$ do not envy any vertex of $V(G_t)$, while also keeping track of the number of unoriented edges of the partial orientation as well as the following information for each of the at most $\mathrm{tw}+1$ vertices in B_t:

- its *revenue* due to edges shared with vertices in $V(G_t) \setminus B_t$,
- its *demand* so that it does not envy any vertex of $V(G_t) \setminus B_t$.

Observe that both the revenue and the demand of a vertex in B_t can be upper-bounded by W, as a vertex of revenue W cannot envy any other vertex, while a vertex may envy another only up to a value of at most W. Lastly, by Observation 3 we can assume that there are no edges of valuation 0 for both endpoints, thus there are at most $2W$ edges shared between any two vertices, implying

that the number of unoriented edges is at most $2W \cdot \binom{n}{2}$. Consequently, the total running time is $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$. □

We obtain the following corollary for polynomially bounded weights, in which case we have $W = (n+m)^{\mathcal{O}(1)}$.

Corollary 15. *There is an algorithm running in time $(n+m)^{\mathcal{O}(\mathrm{tw})}$ for* Min-Charity EF Orientation *with polynomially-bounded weights.*

Remark 16. The algorithm of Corollary 15 matches the lower bound of $n^{o(\mathrm{vc})}$ in Theorem 12. Recall that the treewidth of a graph is upper-bounded by its vertex cover number and that for simple graphs, $m = \mathcal{O}(n^2)$ holds.

4 EFX Orientations

The following observations will be useful throughout this section. Recall the definition of *strong* envy from Definition 1.

Observation 17. *An agent that has only received one good is not strongly envied by any agent.*

Observation 18. *For agents i, j, if i does not strongly envy j and j has received a good g s.t. $v_i(g) = 0$, then i does not envy j.*

4.1 From EF to EFX Orientations

It was first proven by Christodoulou et al. [23] that EFX Orientation is NP-complete even with symmetric valuations; this hardness persists even for simple graphs of vertex cover number 8 or for multigraphs consisting of 10 vertices, where the problem was proven to be weakly NP-complete (Deligkas et al. [27]). In a follow-up work, Afshinmehr et al. [2] show NP-completeness even for symmetric instances on bipartite multigraphs with 8 vertices. Here we present a reduction from EF Orientation to EFX Orientation that, along with Section 3.2, will allow us to complement and improve upon the aforementioned results. See Figure 4 for an overview of our reduction.

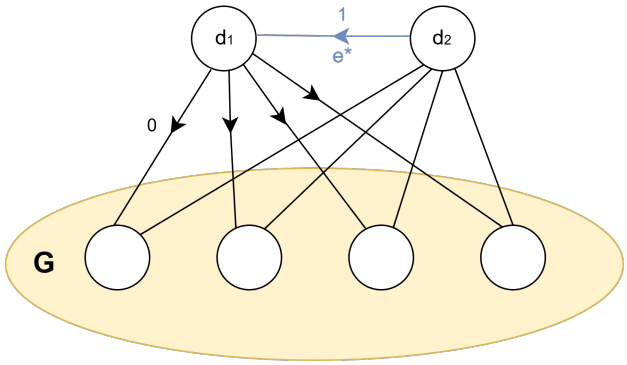

Figure 4: Our reduction from EF Orientation to EFX Orientation in Theorem 19. If e^* is oriented towards d_1, then all other edges incident to d_1 must be oriented away from d_1 to obtain an EFX orientation. Every $v \in G$ receives an edge of weight 0, which forces equivalence between EF and EFX by Observation 18. A symmetrical argument holds if e^* is oriented towards d_2. For more details, see Appendix C.

Theorem 19 (★). *There is a polynomial-time, logarithmic-space reduction from* EF Orientation *to* EFX Orientation *that only adds 2 vertices and edges of symmetric binary valuation. The reduction preserves simplicity.*

Combining Theorem 19 with the hardness results in Section 3.2 leads to Corollaries 20 and 21.

Corollary 20. EFX Orientation *is weakly NP-complete even restricted to*

- *simple symmetric instances of vertex cover number* 4*,*
- *symmetric instances with* 4 *vertices.*

Corollary 21. EFX Orientation *is W[1]-hard parameterized by vertex cover number, XNLP-hard parameterized by pathwidth, and XALP-hard parameterized by treewidth, even restricted to simple symmetric instances with polynomially-bounded weights. Furthermore, for such instances,* EFX Orientation *does not admit a* $n^{o(\mathrm{vc})}$ *algorithm under the ETH, where* vc *denotes the vertex cover number of the input graph.*

We now compare Corollaries 20 and 21 with existing literature. Corollary 20 improves upon the respective constants of 8 and 10 from Deligkas et al. [27], as well as the result for multigraphs with 8 vertices by Afshinmehr et al. [2]. As for Corollary 21, it presents the first hardness results for instances with polynomially-bounded weights and small vertex cover, and along with the algorithm of Section 4.2, it answers an open question of Deligkas et al. [27] regarding the hardness of EFX Orientation in this setting.

4.2 Parameterization by Treewidth plus Maximum Shared Weight

Moving on, we consider the EFX Orientation problem on graphs of small treewidth. As previously mentioned, if one allows binary-encoded weights, then the problem remains NP-hard even for simple graphs of constant vertex cover, or multigraphs with constant number of vertices. Consequently, in the following we consider the case where all weights are polynomially-bounded; this is exactly the regime for which Deligkas et al. [27] have explicitly asked whether the structure of the input graph may prove algorithmically useful. For this case, Corollary 21 already gives a $n^{o(\mathrm{vc})}$ lower bound under the ETH even on simple graphs. Our main contribution in this section is to complement this lower bound with a DP algorithm parameterized by treewidth plus maximum shared weight, in a similar fashion as in Theorem 14; for polynomially-bounded weights where $W = (n+m)^{\mathcal{O}(1)}$ this results in an algorithm of running time $(n+m)^{\mathcal{O}(\mathrm{tw})}$. Again, our algorithm works even for the more general Min-Charity EFX Orientation problem, and for multigraphs with non-identical valuations.

We first state the following observation.

Observation 22. *Let an instance of* EFX Orientation *where vertices* i, j *share more than one edges of weight* 0 *for both endpoints. Then, removing all but one of those edges results in an equivalent instance.*

To see why Observation 22 holds, notice that any EFX orientation that does not orient all such edges to a *single* endpoint, can be modified to an EFX orientation that does.

Theorem 23 (★). *Given an instance* $\mathcal{I}$ *of* Min-Charity EFX Orientation *along with a nice tree decomposition of the input multigraph of width* tw*, one can decide* $\mathcal{I}$ *in time* $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$*, where* W *denotes the maximum shared weight of* $\mathcal{I}$ *and* n *the number of agents.*

Proof sketch. The algorithm closely follows the one of Theorem 14. For a node t of the given nice tree decomposition, let G_t denote the corresponding subgraph, while B_t denotes the set of vertices appearing in the bag of node t. Once again, for every node t of the tree decomposition, we keep track of whether there exists a partial orientation of G_t (excluding the edges among vertices of B_t) that guarantees that the vertices of $V(G_t) \setminus B_t$ do not strongly envy the vertices of $V(G_t)$, while also keeping track of the number of unoriented edges of the partial orientation as well as the following information for each of the at most $\mathrm{tw} + 1$ vertices in B_t:

- its *revenue* due to edges shared with vertices in $V(G_t) \setminus B_t$, separately handling the case where a vertex has no received goods,
- its *demand* so that it does not strongly envy any vertex of $V(G_t) \setminus B_t$, separately handling the case where a vertex is non-strongly envied by a vertex of $V(G_t) \setminus B_t$.

Notice that both the revenue and the demand of a vertex in B_t can be upper-bounded by W, as a vertex with revenue W cannot envy any other vertex, while a vertex may envy another up to a value of at most W. Lastly, by Observation 22 we can assume that for every pair of vertices there is at most one edge of valuation 0 for both endpoints, thus there are at most $2W + 1$ edges shared between them, implying that the number of unoriented edges is at most $(2W+1) \cdot \binom{n}{2}$. Consequently, the total running time is $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$. □

Just like in Section 3.4, we obtain the following.

Corollary 24. *There is an algorithm running in time* $(n+m)^{\mathcal{O}(\mathrm{tw})}$ *for* Min-Charity EFX Orientation *with polynomially-bounded weights.*

Remark 25. The algorithm of Corollary 24 matches the lower bound of $n^{o(\mathrm{vc})}$ in Corollary 21.

5 Conclusion

In this work we initiated the study of EF orientations in multigraphs and presented a variety of both positive and negative results regarding their computational complexity, including a complete characterization for the case of binary valuations. We additionally studied them under the perspective of parameterized complexity, and showed how some of the results can even be transferred to the related EFX Orientation problem, improving upon previous work on the latter in the process and settling an open question by Deligkas et al. [27]. Our work demonstrates that EF orientations are both interesting in their own right, and useful for transferring results to EFX orientations.

As a direction for future work, we comment that Theorem 13 generalizes Theorem 6, albeit only for simple graphs. A natural question is thus whether we can develop such an algorithm that works for multigraphs as well, that is, whether EF Orientation admits an FPT algorithm parameterized by the number k of heavy edges running in time $2^k n^{\mathcal{O}(1)}$. Another interesting direction concerns our parameterized algorithms in Sections 3.4 and 4.2, specifically, whether our parameterization by treewidth plus maximum shared weight can be improved to one by treewidth plus *maximum weight*.

One last future research direction we would like to highlight is the search for regimes in which the complexity of EF and EFX Orientation differs. We showed that one such case regards instances with binary valuations, while the two problems have similar behavior in the case of polynomially-bounded weights and small vertex cover/treewidth. In which other (valuation or structural) regime does the complexity of the two problems differ?

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A Parameterized Complexity

In this section we briefly review some basic notions of parameterized complexity which we use throughout the paper; for a more thorough exposition we refer to standard textbooks, e.g., [24, 29].

An instance of a *parameterized problem* is a pair (x, k), where x is the main input (with size $n := |x|$) and $k \in \mathbb{N}$ is the parameter. We say that a parameterized problem is *fixed-parameter tractable* (FPT) and belongs to the class FPT if it admits an algorithm of running time $f(k) \cdot n^{\mathcal{O}(1)}$ for some computable function f.

Let Π_1 and Π_2 be two parameterized problems. A *parameterized reduction* (also known as an *FPT reduction*) from Π_1 to Π_2 is an algorithm that, given an instance (x, k) of Π_1, produces an instance (x', k') of Π_2 such that:

- (x, k) is a yes-instance of Π_1 if and only if (x', k') is a yes-instance of Π_2,
- $k' \leq g(k)$ for some computable function g,
- the running time of the algorithm is $f(k) \cdot |x|^{\mathcal{O}(1)}$ for some computable function f.

Notice that if Π_2 is fixed-parameter tractable (FPT), then so is Π_1.

We will use parameterized reductions to show hardness results for parameterized problems. In particular, we define the class W[1] as the set of parameterized problems that are FPT-reducible to k-Clique parameterized by k.[5] It holds that FPT $\subseteq$ W[1], and it is widely believed that FPT $\neq$ W[1], thus showing that a problem is W[1]-hard is strong evidence against fixed-parameter tractability [24, 29].

Another complexity class considered in this paper is XNLP. This class was recently introduced by [15], and it consists of parameterized problems decidable by a nondeterministic algorithm that, on inputs of size n with parameter k, uses time $f(k) \cdot n^{\mathcal{O}(1)}$ and space $f(k) \cdot \log n$, for some computable function f. It holds that W[1] $\subseteq$ XNLP, and in fact XNLP-hardness implies W[t]-hardness for all $t \in \mathbb{N}$. In order to show XNLP-hardness results, we need to introduce the notion of *parameterized logspace reductions*, which are simply parameterized reductions that use $\mathcal{O}(g(k) + \log |x|)$ space, where g is a computable function. In that case, if Π_1 is XNLP-hard and there is a parameterized logspace reduction from Π_1 to Π_2, then Π_2 is also XNLP-hard [15]. Another class we use is XALP, which can be seen as the "tree variant" of XNLP, where we additionally allow access to an auxiliary stack (with only top element lookup allowed) [14]. This class contains XNLP, and XALP-hardness can be shown via parameterized logspace reductions as well. XNLP and XALP have been shown to be the "natural home" of many problems parameterized by pathwidth and treewidth respectively, as many problems tend to be complete for each class respectively.

The *treewidth* of a graph is a measure of how close the graph is to being a tree, and is arguably the most well-studied structural parameter. A *tree-decomposition* $\mathcal{T}$ of a graph G is a pair $(T, \{B_t \mid t \in V(T)\})$ where T is a tree and each node t of T is assigned a *bag* $B_t \subseteq V(G)$, such that: (i) every vertex $v \in V(G)$ is contained in some bag B_t; (ii) for every edge $\{u, v\} \in E(G)$ there exists $t \in V(T)$ with $\{u, v\} \subseteq B_t$; and (iii) for every $v \in V(G)$, the set $\{t \in V(T) \mid v \in B_t\}$ induces a connected subtree of T. The *width* is $\max_{t \in V(T)} |B_t| - 1$; the *treewidth* $\mathrm{tw}(G)$ is the minimum width over all tree-decompositions of G. A tree-decomposition is *nice* if each node of T has one of the following types: (i) *Leaf* nodes t where $B_t = \varnothing$; (ii) *Introduce* nodes t with a unique child t' such that $B_t = B_{t'} \cup \{v\}$

[5] The k-Clique problem asks, given a graph G and an integer k, whether G contains a clique of size at least k.

for some $v \in V(G)$; (iii) *Forget* nodes t with a unique child t' such that $B_{t'} = B_t \cup \{v\}$ for some $v \in V(G)$; (iv) *Join* nodes t with exactly two children t_1, t_2 such that $B_t = B_{t_1} = B_{t_2}$. It is known that given any tree decomposition one can in linear time construct a nice tree decomposition of the same width. Computing a tree-decomposition of minimum width is in FPT when parameterized by the treewidth [11, 44, 46], and even more efficient algorithms exist for obtaining near-optimal tree-decompositions [45].

We conclude this section by recalling two well-known hypotheses from the area of fine-grained complexity [39, 40]. The *Exponential Time Hypothesis* (ETH) posits that there exists a constant $c > 0$ such that no algorithm solves 3-SAT on n variables and m clauses in time $2^{c(n+m)} \cdot (n+m)^{\mathcal{O}(1)}$. The *Strong Exponential Time Hypothesis* (SETH) states that for every $\varepsilon > 0$ there exists a constant $k \geq 3$ (depending only on ε) such that no algorithm solves k-SAT on n variables in time $(2-\varepsilon)^n \cdot n^{\mathcal{O}(1)}$. Both hypotheses constitute standard tools to obtain oftentimes tight (conditional) lower bounds. With respect to parameterized complexity, they have been used to pinpoint the best parametric dependence possible for various parameterized algorithms [48, 49].

B Proofs for Section 3 (EF Orientations)

Theorem 5 (Characterization ★)**.** *Let $\mathcal{I} = (G = (V, E), \{v_i\}_{i \in V})$ be an instance of* EF Orientation *with binary valuations and define $G' = (V, E')$ as the subgraph of G s.t. $E' \subseteq E$ consists only of the edges in E with weight 1 for both their endpoints. $\mathcal{I}$ has an EF orientation if and only if for every connected component C of G' at least one of the following properties holds.*

1. *C contains a circuit[6] consisting of at least 3 vertices.*
2. *C contains a vertex i incident with some edge $e \in E_{ij}$ s.t. $v_i(e) = 1$ and $v_j(e) = 0$.*
3. *C contains two vertices i, j s.t. $|E'_{ij}|$ is even and non-zero.*
4. *C consists of only one vertex.*

Proof. We will prove the forward direction of the theorem by constructing an EF orientation given each of the four properties. As already explained, we orient all edges $e \in E_{ij}$ s.t. $v_i(e) = 1$ and $v_j(e) = 0$ towards i (for all $i, j \in V$), and remove all edges with weight 0 for both endpoints. From now on we may assume that all unoriented edges have weight 1 for both of their endpoints (i.e., all unoriented edges are in the set E'), hence G has an EF orientation if and only if every connected component of $G' = (V, E')$ has an EF orientation, granted the aforementioned orientation for all edges in $E \setminus E'$. We will prove that if any of the four properties hold for some connected component C of G', then C has an EF Orientation, by repeatedly applying Observation 4.

Property 1. Let $C = (V_c, E_c)$ be a connected component of G' satisfying Property 1, i.e., C contains a circuit $B = (V_b, E_b)$ with $|V_b| \geq 3$. By the definition of circuit, there exists a cyclic route that visits all edges in E_b exactly once, starting and ending at the same vertex in V_b. Pick such a route and orient all edges in E_b along the route; this way, every vertex in V_b receives at least one edge (see Figure 5). Let i, j be any two vertices in V_b. By the above, it holds that either i or j has now received some edge $e \notin E_{ij}$ with weight 1 (if both i and j had only received edges in E_{ij}, that would imply that $|V_b| = 2$, contradiction). Hence, we can apply Observation 4 to orient all

[6] In graph theory, a circuit is defined as a closed trail where a sequence of vertices starts and ends at the same vertex without repeating any edges. Informally, it is a cycle that may pass through each vertex more than once, while only using each edge at most once.

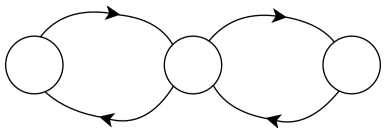

Figure 5: An example of a circuit consisting of 3 vertices, without containing a cycle of length greater than 2. Observe that no vertex is envious of another if the edges are oriented along a cyclic route as shown (assuming all edges have weight 1 for both endpoints).

unoriented edges in E'_{ij} while preserving envy-freeness (for all $i, j \in V_b$). Since every vertex in V_b has received an edge in E_b, we can also apply Observation 4 to orient all edges between vertices in V_b and their neighbors that are not in V_b, which forces the neighbors of the vertices in V_b to receive at least one edge of weight 1. This renders Observation 4 applicable to orient the edges between those vertices and their own neighbors, and so on. This orients all $e \in E_c$ while preserving envy-freeness.

Property 2. Let $C = (V_c, E_c)$ be a connected component of G' satisfying Property 2, i.e., there is a vertex $i \in V_c$ that has received some edge $e \in E_{ij}$ s.t. $v_i(e) = 1$ and $v_j(e) = 0$, for some $j \in V$ (see Figure 6). This renders Observation 4 applicable[7] to orient all edges incident with i, and then we can apply it iteratively to orient the edges incident with the neighbors of i, and so on, as we did for Property 1, thus orienting all $e \in E_c$ while preserving envy-freeness.

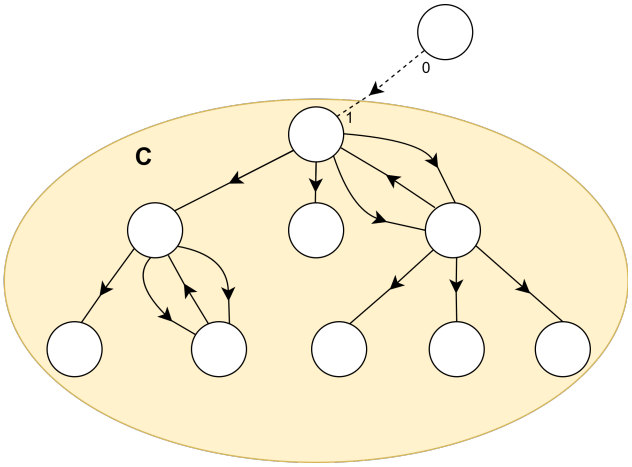

Figure 6: An example of a connected component C of G' that satisfies Property 2 of Theorem 5, i.e., some vertex receives an edge of weight 1 for itself and weight 0 for the other endpoint. This allows us to repeatedly apply Observation 4 starting from that vertex, orienting all edges of C to obtain an EF orientation as demonstrated.

Property 3. Let $C = (V_c, E_c)$ be a connected component of G' satisfying Property 3, i.e., there are vertices $i, j \in V_c$ s.t. $|E'_{ij}| > 0$ is even. We orient half of these edges towards i and the other half towards j, thus ensuring that i and j do not envy each other and each of them has received an edge of weight 1. Now we can apply Observation 4 to orient the unoriented edges between i and its neighbors, and then their own neighbors, and so on (same for j and its neighbors), as we did for Property 1. This orients all $e \in E_c$ while preserving envy-freeness.

Property 4. In this case, the empty set is a trivial EF orientation for C.

[7] Since $v_j(e) = 0$, Observation 4 is applicable for all neighbors of i, including j.

This concludes the forward direction of the proof. It remains to prove the converse, i.e., that if for some connected component $C = (V_c, E_c)$ of G' none of the four properties hold, then C has no EF orientation.

Converse. Let $C = (V_c, E_c)$ be a (multi)graph that violates all four properties. Then:

- C is a tree that allows an odd[8] number of parallel edges between neighbors (instead of just one).
- For every $i \in V_c$ there is at most one[9] vertex $j \in V_c$ s.t. $|E_{ij}| > 1$.
- For every $i \in V_c$ and every $j \in V$ (note that j does not have to be in V_c) there is no $e \in E_{ij}$ s.t. $v_i(e) = 1$ and $v_j(e) = 0$. Thus, every edge in E incident to some vertex in V_c has weight 1 for both endpoints.
- $|E_c| \geq 1$.

Fix an arbitrary root r for the tree C and let $L_1 = \{r\}$ be the first *level* of the tree, let L_2 be the second level consisting of the neighbors of r, and so on, up to level $L_{\max}$. For $v \in V_c$ we define $s(v)$ as the number of edges connecting v to the previous level of the tree. Note that, by the assumptions for C, each $v \in V$ (excluding the root r) is connected to exactly one vertex of the previous level, with an odd amount of edges. We will prove the following claim through induction on the levels L_i.

Claim 26. *Every $v \in L_i$ must receive at least $(s(v) - 1)/2 + 1$ of the $s(v)$ edges connecting it to a vertex of level L_{i-1}, for all $i = \{2, \ldots, \max\}$.*

Proof. Induction basis. Let $i \in L_{\max}$ be a leaf and $j \in L_{\max -1}$ be its (only) neighbor. If $|E_{ij}| = 1$, then it is clear that the edge between i and j must be oriented towards i in any EF orientation (otherwise, i would envy j). Similarly, if $|E_{ij}| > 1$, then $|E_{ij}|$ must be odd and at least $(|E_{ij}|-1)/2+1$ edges must be oriented towards i.

Induction hypothesis. Suppose that every $v \in L_{k+1}$ $(k \geq 2)$ must receive at least $(s(v) - 1)/2 + 1$ of the $s(v)$ edges connecting it to a vertex of level L_k.

Inductive step. We now prove that every $i \in L_k$ of the tree must receive at least $(s(i) - 1)/2 + 1$ of the $s(i)$ edges connecting it to a vertex of level L_{k-1}. Consider three cases:

- Case 1: $|E_{ij}| \leq 1$ for all $j \in V_c$. By induction hypothesis, i receives none of the edges connecting it to level L_{k+1}, hence it must receive the (only) edge connecting it to level L_{k-1}, as it would otherwise envy its neighbors in L_{k+1}.
- Case 2: $|E_{ij}| > 1$ for some $j \in L_{k+1}$. By induction hypothesis, j must receive at least $(|E_{ij}| - 1)/2 + 1$ of the edges in E_{ij}, hence i may receive (at most) $(|E_{ij}| - 1)/2$ of these edges. By the assumptions for C, it holds that $|E_{iv}| \leq 1$, $\forall v \neq j,\ v \in V_c$, hence i receives no other edges between itself and level L_{k+1}, and is connected to level L_{k-1} by a single edge. This edge must be oriented towards i, as otherwise i would envy j.
- Case 3: $|E_{ij}| > 1$ for some $j \in L_{k-1}$. By the assumptions for C, it holds that $|E_{iv}| \leq 1$, $\forall v \neq j,\ v \in V_c$. Thus, by induction hypothesis, i receives none of the edges between itself and

[8] Non-zero even numbers of parallel edges are not allowed, due to the negation of Property 3.

[9] If some v was connected with two other vertices through a number of edges greater than one, that would induce a circuit consisting of 3 vertices, satisfying Property 1. See Figure 5 for an example.

level L_{k+1}. In order for i to not envy j, at least $(|E_{ij}| - 1)/2 + 1$ of the edges in E_{ij} have to be oriented towards i.

This concludes the proof of the claim. □

We are now ready to prove the converse direction of the theorem. By the assumptions for C, it holds that the root r has at most one neighbor $j \in L_2$ s.t. $|E_{rj}| > 1$. First, suppose such a neighbor j exists. By Claim 26, at least $(|E_{rj}| - 1)/2 + 1$ of the edges in E_{rj} must be oriented towards j and all other edges incident to r must be oriented towards the respective vertices in L_2. This causes r to envy j. Now, suppose j did not exist, i.e., $|E_{ri}| \leq 1,\ \forall i \in V_c$. For this case, Claim 26 implies that r cannot receive any edges at all, causing it to envy all of its neighbors in L_2. In both cases, r is envious of some other vertex, hence there is no EF orientation for C (and, thus, neither for G). □

Theorem 6 (★)**.** *There is an algorithm running in time $\mathcal{O}(n + m)$ for* EF Orientation *with binary valuations.*

Proof. Define $G' = (V, E')$ as the subgraph of G s.t. $E' \subseteq E$ consists only of the edges in E with weight 1 for both their endpoints. By Theorem 5, it suffices to check whether every connected component of G' satisfies at least one of the four properties mentioned in that theorem. Properties 2, 3, 4 can be trivially checked in $\mathcal{O}(n + m)$ time for all connected components. For Property 1, we observe that if a multigraph contains a circuit consisting of at least 3 vertices and does not contain a cycle of length at least 3, then there must exist vertices i, j, k s.t. $|E'_{ij}| \geq 2$ and $|E'_{jk}| \geq 2$. Hence, for Property 1 it suffices to run a simple cycle-detection algorithm like depth-first search, and additionally check whether there exists a vertex j with neighbors i, k s.t. $|E'_{ij}| \geq 2$ and $|E'_{jk}| \geq 2$. □

Theorem 7 (★)**.** *There is an algorithm running in time $\mathcal{O}(n + m)$ for* Min-Charity EF Orientation *with binary valuations.*

Proof. Define $G' = (V, E')$ as the subgraph of G s.t. $E' \subseteq E$ consists only of the edges in E with weight 1 for both their endpoints. By Theorem 5, an EF orientation for G exists if and only if every connected component of G' satisfies at least one of the properties mentioned in that theorem. It follows that the answer to Min-Charity EF Orientation is the sum over the minimum amount of edges that must be removed for each connected component to satisfy one of the four properties.

By Theorem 6, we can check whether a connected component satisfies some of the four properties in $\mathcal{O}(n + m)$ time (in which case, the answer for that component is 0). We now assume that some component violates all four properties. If it contains vertices i, j with $E'_{ij} > 1$, then we can remove one edge to satisfy Property 3. It is clear that Properties 1 and 2 can never be satisfied by removing edges from the input; hence, if a connected component violates all four properties and contains no vertices i, j with $|E'_{ij}| > 1$, then our only option is to remove all edges in that component to satisfy Property 4. All of the aforementioned operations can be done in time $\mathcal{O}(n + m)$. □

Theorem 9 (★)**.** *There is no $(2 - \varepsilon)^k n^{\mathcal{O}(1)}$ algorithm for* EF Orientation *under SETH, where k is the number of edges of value at least 2, even when restricted to simple symmetric instances with polynomially-bounded weights.*

Proof. We will reduce SAT to EF Orientation. Let ϕ be an instance of SAT, where $X = \{x_1, \ldots, x_n\}$ denotes its variables and $C = \{c_1, \ldots, c_m\}$ its clauses. We will construct an equivalent

simple symmetric instance $\mathcal{I} = (G, v)$ of EF Orientation. Since the instance is symmetric, we will only keep track of one weight per edge.

Let w_i be equal to the maximum between the number of clauses of C that contain x_i and the number of clauses of C that contain $\neg x_i$; formally, $w_i = \max\{|\{c_j \in C \mid x_i \in c_j\}|, |\{c_j \in C \mid \neg x_i \in c_j\}|\}$. For $i \in [n]$ we introduce vertices t_i and f_i and an edge $\{t_i, f_i\}$ with weight w_i. Informally, the orientation of $\{t_i, f_i\}$ towards t_i corresponds to a truthful assignment of x_i. Furthermore, for all $\ell \in [w_i]$, we introduce vertices t_i^ℓ, f_i^ℓ and edges $\{t_i^\ell, t_i\}$ and $\{f_i^\ell, f_i\}$, all of weight 1.

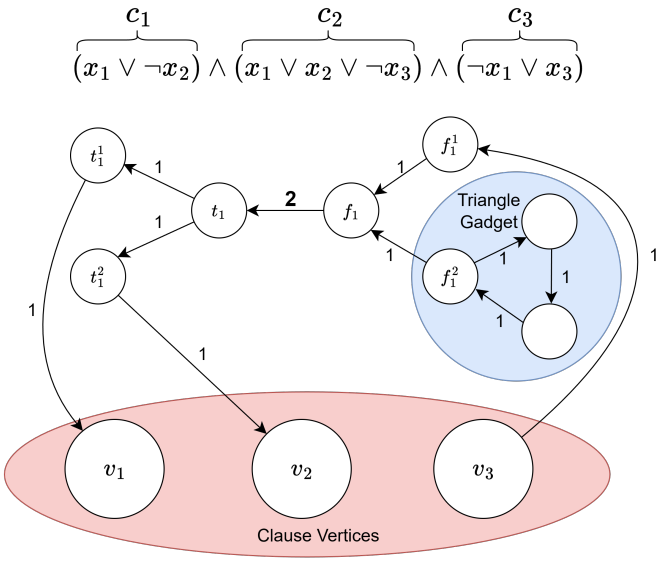

Figure 7: An example of our construction in Theorem 9. For simplicity, we only show the part of the graph corresponding to x_1, for which $w_1 = 2$. Each of the two copies of t_1, f_1 is connected with some clause vertex whose respective clause contains a positive/negative occurrence of x_1 respectively. f_1^2 is *leftover*, and hence attached to a triangle gadget. The orientation shown corresponds to $\alpha(x_1) = \mathsf{true}$. Observe that all non-clause vertices are not envious of their neighbors; additionally, all clause vertices corresponding to clauses satisfied by $\alpha(x_1) = \mathsf{true}$ are not envious of their neighbors, i.e., v_3 is the only envious vertex in the figure. Observe that, if the same construction is repeated for x_3 and $\alpha(x_3) = \mathsf{true}$, then v_3 will also be rendered non-envious.

For each c_j we create a vertex v_j corresponding to that clause; we connect these *clause* vertices with the vertices t_i^ℓ, f_i^ℓ as follows. For $i \in [n]$, we connect t_i^ℓ with the vertex corresponding to the ℓ-th clause that contains a positive occurrence of x_i, with an edge of weight 1. Note that, since the amount of clauses containing a positive occurrence of x_i is at most w_i (by construction), some vertices t_i^ℓ may remain not adjacent to any clause vertex v_j by the aforementioned procedure. For every such *leftover* vertex u that is adjacent to no clause vertex, we create the following *triangle* gadget. We introduce two new vertices $a_{i,\ell}^u$ and $b_{i,\ell}^u$ and the edges $\{u, a_{i,\ell}^u\}$, $\{a_{i,\ell}^u, b_{i,\ell}^u\}$ and $\{b_{i,\ell}^u, u\}$, all of weight 1, thus creating a triangle between u, $a_{i,\ell}^u$ and $b_{i,\ell}^u$. We apply exactly the same construction to vertices f_i^ℓ for *negative* occurrences of x_i.

This concludes the construction of $\mathcal{I}$. For an illustration of the construction we refer to Figure 7. We now prove that ϕ has a satisfying assignment iff $\mathcal{I}$ has an EF orientation.

($\Rightarrow$) Let $\alpha\colon X \to \{\mathsf{true}, \mathsf{false}\}$ be a satisfying assignment for ϕ. We construct an orientation O of G

as follows. For $i \in [n]$ such that $\alpha(x_i) = \mathsf{true}$, we orient the edge $e = \{t_i, f_i\}$ towards t_i. Then, we orient all edges incident to f_i (except e) towards f_i and all edges incident to t_i (except e) away from t_i. Furthermore, any edge (t_i^ℓ, v_j) is oriented towards v_j, and any edge (f_i^ℓ, v_j) is oriented away from v_j. For $i \in [n]$ such that $\alpha(x_i) = \mathsf{false}$, we orient all of the aforementioned edges in the opposite manner. For each triangle gadget on vertices $\{u, a_{i,\ell}^u, b_{i,\ell}^u\}$, orient its edges as a directed 3-cycle (e.g. $u \to a_{i,\ell}^u \to b_{i,\ell}^u \to u$). Thus, each of the three vertices receives some edge of weight 1.

We now prove that O is EF. Let $i \in [n]$ such that $\alpha(x_i) = \mathsf{true}$. Then, t_i has received an edge of weight w_i and f_i has received w_i edges of weight 1 (by construction), hence t_i, f_i do not envy each other. Symmetrically, the same holds for $\alpha(x_i) = \mathsf{false}$. For all vertices in the graph other than t_i, f_i vertices it holds that they are only incident to edges of weight 1. Since the graph is simple, if any vertex (other than t_i, f_i vertices) receives at least one edge, then it is clear that it does not envy any of its neighbors. Observe that every vertex of the graph receives at least one edge; in particular:

- every clause vertex v_j receives at least one edge because α is a satisfying assignment for ϕ (meaning that every clause c_j has at least one satisfied literal, i.e., v_j receives some edge from a vertex t_i^ℓ if the satisfied literal is positive or a vertex f_i^ℓ if the satisfied literal is negative).
- every vertex t_i^ℓ, f_i^ℓ that is connected to some clause vertex receives either the edge connecting it to that clause vertex, or the edge connecting it to t_i or f_i, due to the construction of O.
- every vertex t_i^ℓ, f_i^ℓ not connected to any clause vertex v_j is part of some triangle gadget, and thus receives at least one edge.

Thus, we have proven that no vertex envies another, meaning that O is EF.

($\Leftarrow$) Let O be an EF orientation of G. We define an assignment α by setting $\alpha(x_i) = \mathsf{true}$ iff the edge $\{t_i, f_i\}$ is oriented towards t_i. We will prove that α is a satisfying assignment.

Let c_j be a clause that is *not* satisfied by α, and let v_j be its corresponding clause vertex. We will prove that v_j receives no edge in O. Towards contradiction, let e be an edge oriented towards v_j. Without loss of generality, we assume that $e = \{v_j, t_i^\ell\}$, for some $i \in [n], \ell \in [w_i]$; symmetrical arguments hold if $e = \{v_j, f_i^\ell\}$. Then, $\{t_i^\ell, t_i\}$ must be oriented towards t_i^ℓ, otherwise t_i^ℓ would envy v_j (recall that t_i^ℓ has only two incident edges, both of weight 1). Now, if $\{t_i, f_i\}$ is not oriented towards t_i, then t_i would envy f_i, since only $w_i - 1$ edges of weight 1 remain available to be oriented towards t_i. Hence, $\{t_i, f_i\}$ must be oriented towards t_i; equivalently, $\alpha(x_i) = \mathsf{true}$. However, by construction, x_i occurs positively in clause c_j (because v_j is connected with t_i^ℓ), which implies c_j is satisfied, leading to a contradiction. We infer that, for an unsatisfied clause c_j, its corresponding clause vertex v_j must not receive any edge in O.

Since O is EF and there are no edges of weight 0 in G, every vertex must receive at least one edge in O (otherwise it would envy all of its neighbors). Combining this with the above, no unsatisfied clause c_j exists, hence α is a satisfying assignment.

This concludes the correctness of the reduction. Now, observe that the number of edges in G of weight at least 2 is at most n. Let $k \leq n$ denote said number; by the reduction we described, a $(2-\varepsilon)^k n^{\mathcal{O}(1)}$ algorithm for EF Orientation would result in a $(2-\varepsilon)^n n^{\mathcal{O}(1)}$ algorithm for SAT, which would contradict the SETH. □

Theorem 10 (★). EF Orientation *is NP-complete even when restricted to simple symmetric instances on* 3*-regular graphs and weights in* $\{1, 2\}$.

Proof. It is clear that EF Orientation is contained in NP, thus we only argue about its NP-hardness. Our reduction is roughly the same as the one presented in Theorem 9, albeit slightly simplified due to reducing from a more appropriate SAT variant.

The problem 2P2N-3SAT is a restricted version of 3-SAT: an instance of 2P2N-3SAT consists of a set X of n variables and a set C of m clauses such that each clause has exactly three literals corresponding to three different variables and each variable appears exactly twice positively and exactly twice negatively. It is known that 2P2N-3SAT is NP-complete [8, 26].

For the construction, we closely follow the one of Theorem 9. For every variable x_i we introduce vertices t_i and f_i connected by an edge of value 2. For every clause c_j we introduce a clause vertex v_j. Then, if $x_i \in c_j$ (resp. $\neg x_i \in c_j$) we connect v_j with t_i (resp. f_i) and set the value of said edge to 1. This completes the construction, and it is easy to see that the graph is simple. Since every literal appears exactly twice and every clause contains exactly 3 literals, it follows that all vertices in the constructed graph are of degree 3.

For the correctness, we proceed as in Theorem 9 and we omit further details due to the similarity of the arguments. □

Theorem 11 (★). EF Orientation *is weakly NP-complete even restricted to*

- *simple symmetric instances of vertex cover number* 2*,*
- *symmetric instances with* 2 *vertices.*

Proof. It is straightforward to verify inclusion in NP. Our reduction follows along the lines of [27, Theorem 3], reducing from the well-known weakly NP-hard Partition problem which asks, given a multiset of positive integers, whether we can partition them into two subsets of equal subset sum.

Let $S = \{s_1, \dots, s_n\}$ be a multiset of n positive integers, such that $\sum_{i=1}^{n} s_i = 2B$ for some $B \in \mathbb{N}$. We construct a symmetric instance $\mathcal{J} = (G, v)$ of EF Orientation (equivalently, we construct an edge-weighted graph G). First, we introduce vertices u_1, u_2, ℓ_1, ℓ_2, and the edges $\{u_1, \ell_1\}$ and $\{u_2, \ell_2\}$, both of weight B. Next, for all $i \in [n]$ we introduce a vertex v_i and connect it with both u_1 and u_2, with both edges being of weight s_i. This concludes the construction, and it is easy to see that $G - \{u_1, u_2\}$ is an independent set, while G is a simple graph.

For the forward direction, let S_1, S_2 denote a partition of S with $\sum_{s \in S_1} s = \sum_{s \in S_2} s = B$. If $v_i \in S_1$, then we orient $\{v_i, u_1\}$ towards u_1 and $\{v_i, u_2\}$ towards v_i. Symmetrically, if $v_i \in S_2$, then we orient $\{v_i, u_1\}$ towards v_i and $\{v_i, u_2\}$ towards u_2. Lastly, for $j \in \{1, 2\}$ we orient $\{u_j, \ell_j\}$ towards ℓ_j. It is straightforward to verify that this is an envy-free orientation.

For the converse direction, let $O = \{X_v\}_{v \in V(G)}$ be an envy-free orientation of $\mathcal{J}$. Let $j \in \{1, 2\}$ and notice that since ℓ_j is a leaf, the edge $\{u_j, \ell_j\}$ is oriented towards ℓ_j in O. Since this edge is of weight B and O is envy-free, it follows that edges of total weight at least B are oriented towards u_j in O.

We argue that for all $i \in [n]$ it holds that $|X_{v_i}| = 1$. If $X_{v_i} = \varnothing$, v_i is envious of both u_1 and u_2, a contradiction. Thus, for all $i \in [n]$, $|X_{v_i}| \geq 1$. Now assume there exists $i \in [n]$ with $|X_{v_i}| = 2$. In that case, it follows that the sum of the values of the edges that are oriented towards either u_1 or u_2 in O is at most $2B - s_i$, contradicting the fact that each of u_1 and u_2 receives weight at least B.

Let $S_j \subseteq S$ such that for all $v_i \in S_j$, the edge $\{v_i, u_j\}$ is oriented towards u_j in O. Due to the previous paragraph, the sets S_1, S_2 partition S, while each of u_1, u_2 receives goods of total value

at least B, while both of them receive goods of total value exactly $2B$. It follows that they both receive goods of value exactly B, and $\sum_{s\in S_1} = \sum_{s\in S_2} = B$.

One can easily modify the previous construction to obtain a multigraph consisting of 2 vertices: we only introduce the vertices u_1, u_2, and for all $i \in [n]$, we introduce an edge of weight s_i connecting them. Correctness is straightforward and we omit the details. □

Theorem 12 (★). EF Orientation *is W[1]-hard parameterized by vertex cover number, XNLP-hard parameterized by pathwidth, and XALP-hard parameterized by treewidth, even restricted to simple symmetric instances with polynomially-bounded weights. Furthermore, for such instances,* EF Orientation *does not admit a* $n^{o(\mathrm{vc})}$ *algorithm under the ETH, where* vc *denotes the vertex cover number of the input graph.*

Proof. The Target Outdegree Orientation problem asks, given a simple graph $G = (V, E)$ with a valuation function $v\colon E \to \mathbb{N}$ and a capacity function $c\colon V \to \mathbb{N}$, whether there exists an orientation $O = \{X_i\}_{i\in V}$ of E such that for all $u \in V$ it holds that $v(X_u) = c$.

It is known that Target Outdegree Orientation is W[1]-hard parameterized by vertex cover, XNLP-hard by pathwidth, and XALP-hard by treewidth, even when all values are given in unary [12, 16]. Furthermore, in [13, Theorem 6.4] (the full version of [12]), the authors give a reduction from Unary Bin Packing to Target Outdegree Orientation such that, in the produced instance, the vertex cover number equals the number of bins, and all weights are encoded in unary. It was recently shown that Unary Bin Packing admits no algorithm running in time $n^{o(k)}$ under the ETH, where k denotes the number of bins [17], improving upon a previous lower bound of $n^{o(k/\log k)}$ [41]; consequently, this results in a $n^{o(\mathrm{vc})}$ ETH lower bound for Target Outdegree Orientation with unary weights, where vc denotes the vertex cover number of the resulting graph.

We present a reduction from Target Outdegree Orientation. Let $\mathcal{I} = (G, v, c)$ denote an instance of said problem. Assume without loss of generality that for all $\{u, v\} \in E$ it holds that $v(\{u, v\}) \leq \min\{c(u), c(v)\}$, as otherwise either this is trivially a No instance, or the orientation of said edge would be forced and we could remove it by updating the capacity of its endpoints and obtain an equivalent instance. Furthermore, we assume that

$$\sum_{e\in E} v(e) = \sum_{u\in V} c(u), \tag{1}$$

as otherwise $\mathcal{I}$ is trivially a No instance.

We construct a symmetric instance $\mathcal{J} = (G' = (V', E'), v')$ of EF Orientation such that G is a simple graph. To obtain G', we extend the graph G by first adding a universal[10] vertex a, and then attaching a leaf b to a. As for the valuation function, we set $v'(e) = v(e)$ for all $e \in E$, $v'(\{a, u\}) = c(u)$ for all $u \in V$, and $v'(\{a, b\}) = \sum_{u\in V} c(u)$. This completes the construction of $\mathcal{J}$, and it is easy to see that deleting a from G' removes any edges added by the construction (thus $\mathrm{vc}(G') \leq \mathrm{vc}(G) + 1$), while the reduction requires logarithmic space. Furthermore, all edges in the constructed instance are of symmetric valuation, while the weight of any introduced edge is polynomially-bounded.

Claim 27. *If* $\mathcal{I}$ *is a Yes instance of* Target Outdegree Orientation*, then* $\mathcal{J}$ *is a Yes instance of* EF Orientation*.*

[10] A vertex is called *universal* if it is adjacent to all other vertices of the graph.

Proof. Let $O = \{X_u\}_{u \in V}$ be an orientation of E such that for all $u \in V$, $v(X_u) = c(u)$. We extend O by orienting the edges in $E' \setminus E$ as follows to obtain an orientation O' of E'. For all $u \in V$, we orient the edge $\{a, u\}$ towards a, while we orient the edge $\{a, b\}$ towards b.

We now argue that O' is envy-free. Vertex b receives its unique incident edge. Vertex a receives all edges connecting it with vertices in V so it is not envious of any such vertex, while the sum of the values of the edges it receives is exactly $\sum_{u \in V} c(u) = v'(\{a, b\})$, thus it is not envious of b. Finally, each vertex $u \in V$ receives edges of total value $c(u) = v'(\{a, u\})$ so it is not envious of a, and for all $u' \in N_G(u)$ it holds that $v'(\{u, u'\}) \leq c(u)$ thus it not envious of any vertex in V. □

Claim 28. *If $\mathcal{J}$ is a Yes instance of* EF Orientation*, then $\mathcal{I}$ is a Yes instance of* Target Outdegree Orientation*.*

Proof. Let O' be an envy-free orientation of G'. Since b is a leaf, it follows that O' orients the edge $\{a, b\}$ towards b. Furthermore, notice that

$$\sum_{u \in V} v'(\{a, u\}) = v'(\{a, b\}),$$

thus it follows that every edge incident to a apart from $\{a, b\}$ is oriented towards a. Now consider $u \in V$ and notice that since (i) the edge $\{a, u\}$ of value $c(u)$ is oriented towards a, and (ii) u is not envious of a, it follows that the sum of the values of the received edges of u is at least $c(u)$, with all such edges belonging to E. In that case, since this holds over all $u \in V$, Equation (1) implies that for every $u \in V$ the sum of the values of its received edges is *exactly* $c(u)$, and all such edges belong to E. Consequently, it follows that the restriction of O' to E is an orientation $\{X_u\}_{u \in V}$ of E such that for all $u \in V$, $v(X_u) = c(u)$. □

This completes the proof of the theorem. □

Theorem 13 (★). *There is an algorithm for* EF Orientation *on simple graphs running in time $2^k n^{\mathcal{O}(1)}$, where k is the number of edges with value at least 2 for at least one endpoint.*

Proof. Let $\mathcal{I} = (G = (V, E), \{v_i\}_{i \in V})$ be an instance of EF Orientation, where G is a simple graph. Given a partial orientation O of G and a vertex $i \in V$, we denote by

$$\begin{aligned}
\sigma_O^-(i) &= \{e \in E \mid e \text{ is oriented towards } i \text{ in } O\},\\
\sigma_O^+(i) &= \{e \in E \mid e \text{ is oriented outwards from } i \text{ in } O\},\\
\sigma_O^0(i) &= \{e \in E \mid e \text{ is incident with } i \text{ and is not oriented in } O\}.
\end{aligned}$$

By Observation 3, we can first preprocess the instance $\mathcal{I}$ as follows. We delete all edges of value 0 for both endpoints, while, if an edge has value 0 for one endpoint and positive value for the other, we orient it towards the latter. Finally, we branch over all the (at most) 2^k orientations of the unoriented edges that have value at least 2 for at least one endpoint. After fixing one such branch, we obtain a partial orientation O in which every edge $e = \{i, j\}$ that is still unoriented satisfies $v_i(e) = v_j(e) = 1$. We define the *revenue* of i under O as

$$r_O(i) = \sum_{e \in \sigma_O^-(i)} v_i(e),$$

and its *demand* under O as

$$d_O(i) = \begin{cases} \max(\{0\} \cup \{v_i(e) \mid e \in \sigma_O^+(i)\}) & \text{if } \sigma_O^0(i) = \varnothing, \\ \max(\{1\} \cup \{v_i(e) \mid e \in \sigma_O^+(i)\}) & \text{otherwise.} \end{cases}$$

Intuitively, $r_O(i)$ is the value allocated to agent i by O, while, as Claim 29 shows, $d_O(i)$ is the minimum value that agent i must receive in any orientation $O^\star$ of G that extends O in order not to envy any of the other agents.

Claim 29. *Let $O^\star$ be an orientation of G that extends O, and let $i \in V$. Agent i does not envy any of its neighbors in $O^\star$ if and only if $r_{O^\star}(i) \geq d_O(i)$.*

Proof. Observe that $\sigma_O^+(i) \subseteq \sigma_{O^\star}^+(i) \subseteq \sigma_O^+(i) \cup \sigma_O^0(i)$, and that for every $e \in \sigma_O^0(i)$ we have $v_i(e) = 1$. Since G is a simple graph, each neighbor of i shares with i exactly one edge. Moreover, i only values edges incident to it, so i does not envy any other agent in $O^\star$ if and only if $r_{O^\star}(i) \geq v_i(e)$ for all $e \in \sigma_{O^\star}^+(i)$.

If $\sigma_O^0(i) = \varnothing$ the statement follows straightforwardly. Now assume otherwise, that is, $\sigma_O^0(i) \neq \varnothing$. If $\sigma_{O^\star}^+(i) \supsetneq \sigma_O^+(i)$, then once again the statement follows straightforwardly, since there exists $e \in \sigma_{O^\star}^+(i) \cap \sigma_O^0(i)$ such that $v_i(e) = 1$.

It remains to consider the case where $\sigma_{O^\star}^+(i) = \sigma_O^+(i)$ and $\sigma_O^0(i) \neq \varnothing$. It suffices to argue that for all envy-free orientations $O^\star$, $r_{O^\star}(i) \geq 1$. Assume otherwise, and let $e = \{i, j\} \in \sigma_O^0(i)$. It holds that e is oriented towards j in $O^\star$, while $r_{O^\star}(i) < 1 = v_i(e)$, thus i is envious of j, a contradiction. This concludes the proof. □

Given Claim 29, our goal is to extend O to an orientation $O^\star$ of G such that $r_{O^\star}(i) \geq d_O(i)$ for every $i \in V$, or to verify that such an extension does not exist. Observe that for all extensions $O^\star$ of O and all agents i it holds that $r_{O^\star}(i) - r_O(i) \leq |\sigma_O^0(i)|$, since i has exactly $|\sigma_O^0(i)|$ incident unoriented edges in O, each of value 1. Consequently, if there exists $i \in V$ with $d_O(i) - r_O(i) > |\sigma_O^0(i)|$, we can immediately conclude that O cannot be extended to an envy-free orientation and proceed with the next branch. In the following, assume that for all $i \in V$ it holds that $d_O(i) - r_O(i) \leq |\sigma_O^0(i)| \leq n$; the second inequality follows from the fact that G is a simple graph.

Claim 30. *There is an algorithm that determines in time $n^{\mathcal{O}(1)}$ whether there exists an envy-free orientation $O^\star$ of G that extends O. If such an orientation exists, the algorithm returns it.*

Proof. Our overarching strategy is to cast our problem as an instance of Upper Degree-Constrained Graph Orientation (UDCGO) (see, e.g., [25, 31, 32] for this and related problems). In UDCGO we are given an undirected graph $G' = (V', E')$ and a function $u \colon V' \to \mathbb{N}$, and we are asked to determine whether there exists an orientation O' of G' such that for all $i \in V'$, $|\sigma_{O'}^+(i)| \leq u(i)$.

Let $\mathcal{J}$ denote the instance of UDCGO obtained by setting $G' = (V, E')$, where $E' \subseteq E$ is the set of edges of G that are not oriented in O, and by defining $u \colon V \to \mathbb{N}$ as follows, for all $i \in V$,

$$u(i) = \begin{cases} |\sigma_O^0(i)| & \text{if } r_O(i) \geq d_O(i), \\ |\sigma_O^0(i)| - (d_O(i) - r_O(i)) & \text{otherwise.} \end{cases}$$

Intuitively, we search for an orientation O' of G' such that for every agent i with $r_O(i) < d_O(i)$, *at least* $d_O(i) - r_O(i)$ edges out of those in $\sigma_O^0(i)$ are oriented towards them. Equivalently, this

is enforced by bounding the number of edges in $\sigma^+_{O^\star}(i) \cap \sigma^0_O(i)$ by $u(i)$. Since any such edge has value 1, this ensures that $r_{O^\star}(i) \geq d_O(i)$ for all $i \in V$, where $O^\star = O \cup O'$. Therefore, by Claim 29, O can be extended to an envy-free orientation $O^\star$ of G if and only if $\mathcal{J}$ is a yes-instance of UDCGO. Moreover, if O' is an orientation of G' witnessing that $\mathcal{J}$ is a yes-instance, then $O^\star = O \cup O'$ is the corresponding envy-free orientation of G.

Lastly, we invoke the polynomial-time algorithm of [31, Theorem 1] to decide $\mathcal{J}$, which, in the yes-case, also returns the desired orientation O'. □

Armed with Claim 30 we simply iterate over all the at most 2^k partial orientations O of the heavy edges, and for each such O check in polynomial time whether it can be extended to an envy-free orientation of G. This yields a $2^k n^{\mathcal{O}(1)}$-time algorithm, as claimed. □

Theorem 14 (★). *Given an instance $\mathcal{I}$ of* Min-Charity EF Orientation *along with a nice tree decomposition of the input multigraph of width* tw*, one can decide $\mathcal{I}$ in time $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$, where W denotes the maximum shared weight of $\mathcal{I}$ and n the number of vertices.*

Proof. Let $\mathcal{I} = (G, \{v_i\}_{i \in V})$ denote the instance of Min-Charity EF Orientation and $(T, \{B_t\}_{t \in V(T)})$ the given nice tree decomposition of G of width tw.

Notations and Definitions. Before describing our algorithm we first fix some notations. For $t \in V(T)$, we denote by V_t the set of vertices of V appearing in the bags of the subtree of T rooted at t excluding those of B_t; more formally, $V_t := \bigcup\{B_{t'} \mid t' \text{ is a descendant of } t \text{ in } T\} \setminus B_t$. Furthermore, let $E_t := \bigcup_{i \in V_t} E_i$ (note that by property of a tree decomposition, the edges of E_t are exactly the ones of $G[V_t]$ plus the ones between V_t and B_t). Finally, let G_t be the graph defined by $G_t := (V_t \cup B_t, E_t)$. We note that B_t is an independent set in G_t.

Let $t \in V(T)$.

We define A_t to be the set of all possible partial orientations of G_t, and A^{ef}_t the set of all possible partial orientations of G_t where all vertices of V_t are non-envious. Intuitively, those are the partial orientations of G_t that can be completed into an envy-free orientation of G.

Signature definition. Let $O = \{X_i\}_{B_t \cup V_t} \in A_t$. We define the *signature* of O with respect to t as a triple $\sigma = (r, d, k)$ with $r \colon B_t \to [0, W]$, $d \colon B_t \to [0, W]$ and $k \in [0, |E|]$ defined by:

- for all $i \in B_t$, $r(i) = \min\{v_i(X_i), W\}$,
- for all $i \in B_t$, $d(i) = \max_{j \in V_t}\{v_i(X_j)\}$,
- $k = |E_t| - \sum_{i \in B_t \cup V_t} |X_i|$

Let Σ_t denote the set of all possible signatures for node t, and notice that $|\Sigma_t| \leq |E|(W+1)^{2(\mathrm{tw}+1)}$. We define the *signature function* $\mathrm{sgn}_t \colon A_t \to \Sigma_t$ that attributes to each partial orientation of G_t its signature with respect to t.

Take $O \in A^{ef}_t$. Intuitively, $\mathrm{sgn}_t(O) = (r, d, k)$ has two types of information: r and d are the minimum amount of information needed to complete O into an partial envy-free orientation of G: $r(i)$ represents what i already *received* from O, up to W as W is enough to never be envious of any other vertex, while $d(i)$ represents what i *demands*, or what it needs to receive to not be envious of any vertex of V_t. This value cannot be bigger than the maximum shared weight W. k is the number of edges from E_t missing in O, it is the amount of edges given to charity in O.

Signature valuation definition We define the *valuation function* $val_t \colon \Sigma_t \to \{0,1\}$ defined by: for all $\sigma \in \Sigma_t$, $val_t(\sigma) = 1$ if and only if there is $O \in A_t^{ef}$ such that $\mathrm{sgn}_t(O) = \sigma$.

Take t to be the root of T; by definition of a nice tree decomposition, B_t is an empty set, $V_t = V$, $E_t = E$ and $G_t = G$. Therefore, A_t^{ef} is exactly the set of all partial envy-free orientations of G. So, take k such that k is minimal and there are r, d such that $val_t((r,d,k)) = 1$, it is equal to the minimal charity of a partial envy-free orientation of G. So, if we have an efficient way to compute the values of val_t, we can efficiently solve Min-Charity EF Orientation.

In the following, we describe the algorithm. It follows the structure of a classical dynamic programming algorithm over a tree decomposition; for all $t \in V(T)$, it computes the valuation of its signatures from the valuation of the signatures of its children in T.

Take $t \in V(T)$:

Leaf node. If t is a leaf node, then $B_t = \varnothing$, $V_t = \varnothing$ and $E_t = \varnothing$. Therefore, $A_t = A_t^{ef} = \{\varnothing\}$. Take *epy* the only function from $\varnothing$ to $[O, W]$, $val_t((epy, epy, 0)) = 1$ and for all $k \in [1, m]$, $val_t((epy, epy, k)) = 0$.

This can be done in linear time trivially.

Introduce node. If t is an introduce node with a child t' such that $B_t = B_{t'} \cup \{i\}$ for some $i \notin B_{t'}$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 31. *For all $\sigma = (r, d, k) \in \Sigma_t$,*

$$val_t(\sigma) = \begin{cases} 0 & \text{if } r(i) \neq 0 \text{ or } d(i) \neq 0 \\ val_{t'}((r|_{B_{t'}}, d|_{B_{t'}}, k)) & \text{otherwise} \end{cases}$$

Intuitively, we extend $val_{t'}$ to B_t to define val_t.

Proof. We note that $V_t = V_{t'}$. Furthermore, by definition of a tree decomposition, i has no edge in common with V_t, and so no edge incident with i is included in E_t: $E_t = E_{t'}$ and i is an isolated vertex in G_t.

Therefore, let $\sigma = (r, d, k) \in \Sigma_t$. If $d(i) \neq 0$ or $r(i) \neq 0$, then there is no $O \in A_t$ where $\mathrm{sgn}_t(O) = \sigma$.

Otherwise, $d(i) = r(i) = 0$, and as $E_t = E_{t'}$, let $O = \{X_j\}_{j \in B_t \cup V_t} \in A_t$, $X_i = \varnothing$. Let $O' = O \setminus \{X_i\}$, $O' \in A_{t'}$, and $\mathrm{sgn}_t(O) = \sigma$ if and only if $\mathrm{sgn}_{t'}(O') = (r|_{B_{t'}}, d|_{B_{t'}}, k)$.

Finally, $O \in A_t^{ef}$ if and only if it makes all the vertices of V_t non-envious, if and only if O' does too, but as $V_t = V_{t'}$, it holds if and only if $O' \in A_t^{ef}$. Therefore, the lemma holds. □

Concerning the complexity, supposing the values of $val_{t'}$ have already been computed, the calculation of each value of val_t can be done in constant time; therefore the calculation of all the values can be done in time $|\Sigma_t| \leq m(W+1)^{2(\mathrm{tw}+1)}$.

Forget node.

Let $i, j \in V$. By Observation 3 there is no edge $e \in E_{ij}$ with $v_i(e) = v_j(e) = 0$, implying that $|E_{i,j}| \leq 2W$. We define the function $M_{ij} \colon [0, v_i(E_{ij})]^2 \times [0, v_j(E_{ij})]^2 \times [0, 2W] \to \{0,1\}$ so that

$M_{ij}((a_1, a_2), (b_1, b_2), k) = 1$ if there is a partial orientation $O = \{X_i, X_j\}$ of the subgraph $(\{i, j\}, E_{ij})$ such that $v_i(X_i) = a_1$, $v_i(X_j) = a_2$, $v_j(X_j) = b_1$, $v_j(X_i) = b_2$, and $|E_{ij} \setminus (X_i \cup X_j)| = k$. We can compute the values of M_{ij} via simple dynamic programming.

If t is a forget node with a child t' such that $B_{t'} = B_t \cup \{i\}$ for some $i \notin B_t$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 32. *For all $\sigma = (r, d, k) \in \Sigma_t$: $val_t((r, d, k)) = 1$ if and only if there are $(r', d', k') \in \Sigma_{t'}$, $\{(a_j^1, a_j^2, b_j^1, b_j^2, k_j)\}_{j \in B_t} \in ([O, W]^4 \times [0, 2W])^{B_t}$ verifying:*

- $val_{t'}((r', d', k')) = 1$
- *for all $j \in B_t$, $M_{ij}((a_j^1, a_j^2), (b_j^1, b_j^2), k_j) = 1$*
- *for all $j \in B_t$, $r(j) = \min\{r'(j) + b_j^1, W\}$*
- *for all $j \in B_t$, $d(j) = \max\{d'(j), b_j^2\}$*
- $d'(i) \leq \sum_{j \in B_t} a_j^1 + r'(i)$
- *for all $j \in B_t$, $a_j^2 \leq r'(i) + \sum_{j \in B_t} a_i^1$*
- $k = k' + \sum_{j \in B_t} k_j$

Intuitively, a partial orientation of G_t is composed of a partial orientation of $G_{t'}$ plus a partial orientation of the edges between i and B_t, which allows us to use the values of $val_{t'}$ to calculate the values of val_t.

Proof. We note that $V_t = V_t' \cup \{i\}$, and $E_t = E_{t'} \cup \bigcup_{j \in B_t} E_{ij}$. Also, $V_t \cup B_t = V_{t'} \cup B_{t'}$.

In that case, a partial orientation $O \in A_t$ can be seen as a partial orientation $O' \in A_{t'}$ together with a partial orientation O'' of $(B_t', \bigcup_{j \in B_t} E_{i,j})$. In the following, we show that given a partial orientation $O \in A_t$, its signature $\mathrm{sgn}_t(O)$ can be computed using only the following information: the signature of O' the restriction of O to $G_{t'}$, and the orientation of $E_{i,j}$ for all $j \in V_t$ of O.

Let $O = \{X_j\}_{j \in V_t \cup B_t} \in A_t$.

For all $j \in B_t$, let $\alpha_j := E_{ij} \cap X_i$, $\beta_j := E_{ij} \cap X_j$, and $X_j' := X_j \setminus E_{ij}$: α_j is the set edges of E_{ij} received by i while β_j is the set of the ones received by j. X_j' is the set of edges received by j from $G_{t'}$.

Let $X_i' := X_i \setminus (\bigcup_{j \in B_t} E_{ij})$ and for all $\ell \in V_{t'}$, let $X_\ell' := X_\ell$. $O' := \{X_j'\}_{j \in B_{t'} \cup V_{t'}}$ is a partial orientation of $G_{t'}$, the restriction of O to $G_{t'}$, so $O' \in A_{t'}$. Let $\sigma = (r, d, k) = \mathrm{sgn}_t(O)$ and $\sigma' = (r', d', k') = \mathrm{sgn}_{t'}(O')$.

We note that $k = k' + \sum_{j \in B_t} |E_{ij} \setminus (\alpha_j \cup \beta_j)|$.

For all $j \in B_t$, let $a_1^j := v_i(\alpha_j)$, $a_2^j := v_i(\beta_j)$, $b_1^j := v_j(\alpha_j)$ and $b_2^j := v_j(\beta_j)$. By linearity of the value functions, we have: $v_i(X_i) = v_i(X_i') + \sum_{j \in B_t} a_j^1$, and for all $j \in B_t$, $v_j(X_j) = v_j(X_j') + b_1^j$, $v_i(X_j) = v_i(X_j') + a_2^j = a_2^j$ and $v_j(X_i) = v_j(X_i') + b_2^j = b_2^j$.

By definition:

- for all $j \in B_t$, $r(j) = \min\{v_j(X_j), W\}$
- for all $j \in B_t$, $d(j) = \max_{l \in V_t}\{v_j(X_l)\}$

In the following, we will reformulate those two previous equalities:

First, for all $j \in B_t$, $r(j) = \min\{v_j(X_j), W\} = \min\{v_j(X'_j) + b^1_j, W\} = \min\{\min\{v_j(X'_j), W\} + b^1_j, W\} = \min\{r'(j) + b^1_j, W\}$.

Second, for all $j \in B_t$, $\max_{l \in V_t}\{v_j(X_l)\} = \max\{\max_{\ell \in V'_t}\{v_j(X'_l)\}, b^j_2\} = \max\{d'(j), b^j_2\}$.

We obtain the two following formulas: for all $j \in B_t$, $r(j) = \min\{r'(j) + b^1_j, W\}$ and $d(j) = \max\{d'(j), b^j_2\}$.

Furthermore, $O \in A^{ef}_t$ if , by definition, O makes the vertices of V_t non-envious, if and only if it does on V'_t and on $\{i\}$.

First, for all $\ell \in V'_t$, $\ell' \in V_t \cup B_t$, $E_{\ell\ell'} \subseteq E_{t'}$. So, as O' is the restriction of O on $E_{t'}$, O makes the vertices of V'_t non-envious if and only if O' does, and therefore if and only if $O' \in A^{ef}_{t'}$.

Second, i is non-envious if it is non-envious of the vertices of B_t and $V_{t'}$. But, on one side, for all $j \in B_t$, $v_i(X_i) = v_i(X'_i) + \sum_{j \in B_t} a^1_i$ and $v_i(X_j) = a^j_2 \leq W$, so $v_i(X_i) \geq v_i(X_j)$ if and only if $r'(i) + \sum_{j \in B_t} a^1_i \geq a^j_2$, and on the other side, $\max_{\ell \in V_{t'}}\{v_i(X_\ell)\} = \max_{\ell \in V_{t'}}\{v_i(X'_\ell)\} = d'(i)$ and $v_i(X_i) = v_i(X'_i) + \sum_{j \in B_t} a^1_j$, so as $d'(i) \leq W$, $v_i(X_i) \geq \max_{\ell \in V_{t'}}\{v_i(X_\ell)\}$ if and only if $r'(i) + \sum_{j \in B_t} a^1_j \geq d'(i)$.

So, O is envy-free on V_t if and only if O' is envy-free on V'_t, $d'(i) \leq r'(i) + \sum_{j \in B_t} a^1_j$ and for all $j \in B_t$, $r'(i) + \sum_{j \in B_t} a^1_i \geq a^j_2$.

We can deduce the following property: let $\sigma = (r, d, k)$ be a signature of t. Then $val_t(\sigma) = 1$ if and only if there is $\sigma' = (r', d', k')$ a signature of t', $O'' = \{X''_j\}_{j \in B'_t}$ an orientation of $\{B'_t, \bigcup_{j \in B_t} E_{ij}\}$ respecting:

- $val_{t'}(\sigma') = 1$
- for all $j \in B_t$, $r(j) = min\{W, r'(j) + v_j(X''_j)\}$
- for all $j \in B_t$, $d(j) = \max\{d'(j), v_j(X''_i)\}$
- $d'(i) \leq v_i(X''_i) + v_i(X'_i)$
- for all $j \in B_t$, $r'(i) + v_i(X''_i) \geq v_j(X''_i)$
- $k = k' + |E_{ij} \setminus (\bigcup_{j \in B_{t'}} X''_j)|$

Therefore, we can deduce the following recursive formula:

$val_t((r, d, k)) = 1$ if and only if there are $(r', d', k') \in \Sigma_{t'}$, $\{(a^1_j, a^2_j, b^1_j, b^2_j, k_j)\}_{j \in B_t} \in ([O, W]^4 \times [0, 2W])^{B_t}$ verifying:

- $val_{t'}((r', d', k')) = 1$
- for all $j \in B_t$, $M_{ij}((a^1_j, a^2_j), (b^1_j, b^2_j), k_j) = 1$
- for all $j \in B_t$, $r(j) = \min\{r'(j) + b^1_j, W\}$
- for all $j \in B_t$, $d(j) = \max\{d'(j), b^2_j\}$
- $d'(i) \leq \sum_{j \in B_t} a^1_j + r'(i)$
- for all $j \in B_t$, $a^2_j \leq (r'(i) + \sum_{j \in B_t} a^1_i)$

- $k = k' + \sum_{j \in B_t} k_j$

□

Concerning the complexity, supposing the values of $val_{t'}$ have already been computed, there are at most $(W^5)^{\mathcal{O}(tw)}$ different set quadruplets of integers and $m \times W^{\mathcal{O}(\mathrm{tw})}$ different signatures of t' to consider. Therefore, this step can be done in time $m \times W^{\mathcal{O}(tw)} \times W^{\mathcal{O}(tw)} = m \times W^{\mathcal{O}(tw)}$.

Join node. If t is a join node with two children t_1, t_2 such that $B_t = B_{t_1} = B_{t_2}$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 33. *For all $\sigma = (r, d, k) \in \Sigma_t$: $val_t((r, d, k)) = 1$ if and only if there are $\sigma_1 = (r_1, d_1, k_1) \in \Sigma_{t_1}$ and $\sigma_2 = (r_2, d_2, k_2) \in \Sigma_{t_2}$ such that:*

- *$val_{t_1}(\sigma_1) = 1$ and $val_{t_2}(\sigma_2) = 1$*
- *for all $j \in B_t$, $r(j) = \min\{r_1(j) + r_2(j), W\}$*
- *for all $j \in B_t$, $d(j) = \max\{d_1(j), d_2(j)\}$*
- *$k = k_1 + k_2$*

Intuitively, a partial orientation of G_t is a combination of a partial orientation of G_{t_1} and G_{t_2}; therefore, we can combine any possible valid partial orientation of G_{t_1} and G_{t_2} into a valid partial orientation of G_t.

Proof. Note that $V_t = V_{t_1} \cup V_{t_2}$, with $V_{t_1} \cap V_{t_2} = \varnothing$. Similarly, $E_t = E_{t_1} \cup E_{t_2}$, with $E_{t_1} \cap E_{t_2} = \varnothing$.

Let $O = \{X_j\}_{j \in \{B_t \cup V_t\}} \in A_t$; for all $j \in B_{t_1} \cup V_{t_1}$, let $X_j^1 := X_j \cap E_{t_1}$ and for all $j \in B_{t_2} \cup V_{t_2}$, let $X_j^2 := X_j \cap E_{t_2}$.

Let $O_1 := \{X_j\}_{j \in \{B_{t_1} \cup V_{t_1}\}}$ and $O_2 := \{X_j\}_{j \in \{B_{t_2} \cup V_{t_2}\}}$. They are the restrictions of O to G_{t_1} and G_{t_2} respectively: $O_1 \in A_{t_1}$ and $O_2 \in A_{t_2}$.

Let $\sigma = (r, d, k) = \mathrm{sgn}_t(O)$, $\sigma_1 = (r_1, d_1, k_1) = \mathrm{sgn}_{t_1}(O_1)$ and $\sigma_2 = (r_2, d_2, k_2) = \mathrm{sgn}_{t_2}(O_2)$.

for all $j \in B_t$, $r(j) = \min\{v_j(X_j), W\}$ and $d(j) = \max_{l \in V_t}\{v_j(X_l)\}$ by definition. In the following, we will rewrite those two equations:

$$\begin{aligned} r(j) &= \min\{v_j(X_j), W\} \\ &= \min\{v_j(X_j^1) + v_j(X_j^2), W\} \\ &= \min\{r_1(j) + r_2(j), W\} \end{aligned}$$

$$\begin{aligned} d(j) &= \max_{l \in V_t}\{v_j(X_l)\} \\ &= \max\{\max_{l \in V_{t_1}}\{v_j(X_l)\}, \max_{l \in V_{t_2}}\{v_j(X_l)\}\} \\ &= \max\{d_1(j), d_2(j)\} \end{aligned}$$

In the end, we obtain $r(j) = \min\{r_1(j) + r_2(j), W\}$ and $d(j) = \max\{d_1(j), d_2(j)\}$.

Furthermore, O is envy-free on V_t if and only if it is envy-free on V_{t_1} and V_{t_2}, if and only if O_1 is envy-free on V_{t_1} and O_2 is envy-free on V_{t_2}. So, $O \in A_t^{ef}$ if and only if $O_1 \in A_{t_1}^{ef}$ and $A_{t_2}^{ef}$.

We obtain the following recursive formula:

$val_t((r,d,k)) = 1$ if and only if there are $\sigma_1 = (r_1, d_1, k_1) \in \Sigma_{t_1}$ and $\sigma_2 = (r_2, d_2, k_2) \in \Sigma_{t_2}$ such that:

- $val_{t_1}(\sigma_1) = 1$ and $val_{t_2}(\sigma_2) = 1$
- for all $j \in B_t$, $r(j) = \min(r_1(j) + r_2(j), W)$
- for all $j \in B_t$, $d(j) = \max\{d_1(j), d_2(j)\}$
- $k = k_1 + k_2$

□

Concerning the complexity, supposing the values of val_{t_1} and val_{t_2} have already been computed, there are $m \times W^{\mathcal{O}(tw)}$ signatures of t_1 and t_2 to consider. Therefore, this step can be done in time $mW^{\mathcal{O}(tw)} \times mW^{\mathcal{O}(tw)} = m^2W^{\mathcal{O}(tw)}$.

This concludes the description and correctness of our algorithm. We now argue about its complexity: as the nice tree decomposition can be supposed to be of polynomial size and each step can be done in time at most $m^2W^{\mathcal{O}(tw)}$, the algorithm has complexity $W^{\mathcal{O}(tw)}(nm)^{\mathcal{O}(1)}$. □

C Proofs for Section 4 (EFX Orientations)

Theorem 19 (★)**.** *There is a polynomial-time, logarithmic-space reduction from* EF Orientation *to* EFX Orientation *that only adds 2 vertices and edges of symmetric binary valuation. The reduction preserves simplicity.*

Proof. Let $\mathcal{I} = (G = (V, E), \{v_i\}_{i \in V})$ be an instance of EF Orientation. We build an EFX Orientation instance $\mathcal{I}'$ by adding two vertices d_1, d_2 to G, each connected with every $v \in V$ by edges of weight 0 (for both endpoints). Additionally, we connect d_1 and d_2 with an edge e^* of weight 1 (for both endpoints). Let $G' = (V', E')$ be the resulting graph of $\mathcal{I}'$ (see Figure 8).

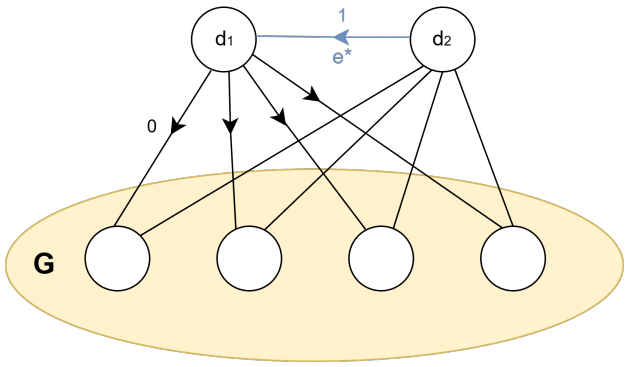

Figure 8: Our reduction from EF Orientation to EFX Orientation. If e^* is oriented towards d_1, then all other edges incident to d_1 must be oriented away from d_1 to obtain an EFX orientation. Every $v \in G$ receives an edge of weight 0, which forces equivalence between EF and EFX by Observation 18. A symmetrical argument holds if e^* is oriented towards d_2.

We prove that if $\mathcal{I}$ has an EF orientation, then $\mathcal{I}'$ has an EFX orientation. Take an EF orientation O of $\mathcal{I}$ and extend it by orienting the edges in $E' \setminus E$ as follows to obtain an orientation O'. Orient e^* towards d_1. Then, orient all other edges incident to d_2 towards d_2 and all other edges incident to d_1 away from d_1. By Observation 17, d_1 is not strongly envied by any $v \in V'$; the same holds for d_2 since all of its allocated edges are of weight 0 (for both endpoints). Additionally, d_1 does not envy any other vertex, since it has received its unique incident edge with non-zero weight. Observe that d_2 only envies d_1, since it has received all of its incident edges except e^* (however, this envy is not strong, by Observation 17). Since O is EF, it holds that u does not envy v for all $u, v \in V$, implying that u does not strongly envy v. From all the above we infer that O' is EFX.

We now prove the converse: if $\mathcal{I}'$ has an EFX orientation, then $\mathcal{I}$ has an EF orientation. Take an EFX orientation O' of $\mathcal{I}'$. Without loss of generality, assume e^* is oriented towards d_1 in O'. Then, for O' to be an EFX orientation, d_1 must not receive any other edge (otherwise d_2 would strongly envy d_1), implying that all other edges incident to d_1 are oriented away from d_1 in O'. Hence, every vertex $v \in V$ has received a zero-weight edge in O'. For all $u, v \in V$, u does not strongly envy v because O' is EFX; by Observation 18, this implies that u does not envy v. Hence, orienting all edges in E in accordance to O' results in an EF orientation O of $\mathcal{I}$.

The reduction described uses logarithmic space,[11] since only two vertices and $\mathcal{O}(n)$ edges of polynomial weight are added to the graph. □

Theorem 23 (★). *Given an instance $\mathcal{I}$ of* Min-Charity EFX Orientation *along with a nice tree decomposition of the input multigraph of width* tw*, one can decide $\mathcal{I}$ in time $W^{\mathcal{O}(\mathrm{tw})}(n+m)^{\mathcal{O}(1)}$, where W denotes the maximum shared weight of $\mathcal{I}$ and n the number of agents.*

Proof. Let $\mathcal{I} = (G, \{v_i\}_{i \in V})$ denote the instance of Min-Charity EF Orientation and $(T, \{B_t\}_{t \in V(T)})$ the given nice tree decomposition of G of width tw.

We will use Observation 22 as well as the following one for the correctness of our algorithm:

Observation 34. *Let O be an orientation. If vertex i envies, but not strongly envies, vertex j, then all edges oriented towards j in O are in E_{ij}. This additionally implies that no vertex $v \neq i$ envies j in O.*

Notations and Definitions. Before describing our algorithm we first fix some notations. For $t \in V(T)$, we define V_t, E_t, G_t and A_t in the same way as in the proof of Theorem 14.

We define A_t^{ef} to be the set of all partial orientations of G_t where, first, all vertices of V_t are non strongly envious, and second, all vertices of B_t which are envied (but not strongly) by some vertex of V_t, are also non strongly envious. Intuitively, those are the partial orientations of G_t that can be completed into an envy-free orientation up to any good of G.

Signature definition. Let $O \in A_t$. We define the *signature* of O with respect to t as a triple $\sigma = (r, d, k)$ with $r \colon B_t \to [0, W] \cup \{None\}$, $d \colon B_t \to [0, W] \cup \{Some\}$ and $k \in [0, |E|]$ defined by:

- for all $i \in B_t$, $r(i) = \begin{cases} None & if\ X_i = \varnothing \\ \min\{v_i(X_i), W\} & otherwise \end{cases}$

[11] This property is essential for the XNLP- and XALP-hardness in Corollary 21.

- for all $i \in B_t$, $d(i) = \begin{cases} Some & if\ there\ is\ \ell \in V_t\ envious\ of\ i \\ \max_{\ell \in V_t,\ e \in X_\ell}\{v_i(X_\ell \setminus \{e\})\} & otherwise \end{cases}$
- $k = |E_t| - \Sigma_{i \in B_t \cup V_t}|X_i|$

Let Σ_t denote the set of all possible signatures for node t, and notice that $|\Sigma_t| \leq |E|(W+2)^{2(tw+1)}$. We define the *signature function* $\text{sgn}_t \colon A_t \to \Sigma_t$ that attributes to each partial orientation of G_t its signature with respect to t.

Take $O \in A_t^{ef}$. As in the algorithm of Theorem 14, $sgn_t(O) = (r, d, k)$ has two types of information: r and d are the minimum amount of information needed to complete O into a partial envy-free orientation up to any good: $r(i)$ represents what i already *received* from O up to any good, and if $r(i) = None$, it means that i did not receive any edge yet, and therefore from Observation 17, it means that in an extension of O, one vertex of $V \setminus V_t$ can be envious of i without being strongly envious (note that we can have $r(i) = 0$, as edges can have a value of 0). $d(i)$ represents the amount that i needs to receive to not be strongly envious of the vertices of V_t, and if $d(i) = Some$, it means that some vertex $\ell \in V_t$ is envious of i (but not strongly envious), and therefore i cannot receive any other edge, as otherwise ℓ would become strongly envious of i.

In the following, we will allow ourselves to use the value $r(i)$ in additions and inequalities, even if it is possible that $r(i) = None$. In that case, we consider that $None$ has value 0.

Signature valuation definition. We define the *valuation function* $val_t \colon \Sigma_t \to \{0, 1\}$ defined by: for all $\sigma \in \Sigma_t$, $val_t(\sigma) = 1$ if and only if there is $O \in A_t^{ef}$ such that $\text{sgn}_t(O) = \sigma$.

Take t to be the root of T; by definition of a nice tree decomposition, B_t is an empty set, $V_t = V$, $E_t = E$ and $G_t = G$. Therefore, A_t^{ef} is exactly the set of all partial envy-free up to any good orientations of G. So, if we have an efficient way to compute the values of val_t, we can efficiently solve Min-Charity EFX Orientation. We present an algorithm that computes the values of val_t for all $t \in V(T)$ dynamically over the nice tree decomposition.

In the following, we describe the algorithm.

Take $t \in V(T)$:

Leaf node. If t is a leaf node, then $B_t = \varnothing$, $V_t = \varnothing$ and $E_t = \varnothing$. Therefore, $A_t = A_t^{ef} = \{\varnothing\}$. Take *epy* the only function from $\varnothing$ to $[0, W]$, $val_t((epy, epy, 0)) = 1$ and for all $k \in [1, m]$, $val_t((epy, epy, k)) = 0$.

Introduce node. If t is an introduce node with a child t' such that $B_t = B_{t'} \cup \{i\}$ for some $i \notin B_{t'}$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 35. *For all $\sigma = (r, d, k) \in \Sigma_t$,*

$$val_t(\sigma) = \begin{cases} 0 & if\ r(i) \neq None\ or\ d(i) \neq 0 \\ val_{t'}((r|_{B_{t'}}, d|_{B_{t'}}, k)) & otherwise \end{cases}$$

Intuitively, we extend $val_{t'}$ to B_t to define val_t.

Proof. We note that $V_t = V_{t'}$. Furthermore, by definition of a tree decomposition, i has no edge in common with V_t, and so no edge incident with i is included in E_t: $E_t = E_{t'}$ and i is an isolated vertex in G_t.

Therefore, let $\sigma = (r, d, k) \in \Sigma_t$. If $d(i) \neq 0$ or $r(i) \neq None$, then there is no $O \in A_t$ where $sgn_t(O) = \sigma$.

Otherwise, $d(i) = 0$, $r(i) = None$ and as $E_t = E'_t$, let $O = \{X_j\}_{j \in B_t \cup V_t} \in A_t$, $X_i = \varnothing$. Let $O' = O \setminus \{X_i\}$, $O' \in A_{t'}$, and $\text{sgn}_t(O) = \sigma$ if and only if $sgn_{t'}(O') = (r|_{B_{t'}}, d|_{B_{t'}}, k)$.

Finally, $O \in A_t^{ef}$ if and only if it makes all the vertices of V_t and all the vertices of B_t envied by vertices of V_t non strongly envious; but, as i is an isolated vertex of G_t, and $V_t = V_{t'}$, it is true if and only if O' makes all the vertices of $V_{t'}$ and all the vertices of $B_{t'}$ envied by vertices of $V_{t'}$ non strongly envious, if and only if $O' \in A_t^{ef}$. Therefore, the lemma holds. □

Concerning the complexity, supposing the values of $val_{t'}$ have already been computed, the calculation of each value of val_t can be done in constant time; therefore the calculation of all the values can be done in time $|\Sigma_t| \leq m(W+2)^{2(tw+1)}$.

Forget node.

Let $i, j \in V$. From Observation 22, we can suppose that $|E_{ij}| \leq 2W + 1$. We define the function $L_{ij}^i \colon [0, W]^2 \times [0, W]^2 \times [0, W] \cup \{\infty\} \times [0, 2W+1] \to \{0, 1\}$ in the following way: $L((a_1, a_2), (b_1, b_2), c, k) = 1$ if and only if there is a partial orientation $O = \{X_i, X_j\}$ of the subgraph $(\{i, j\}, E_{ij})$ such that $v_i(X_i) = a_1$, $v_i(X_j) = a_2$, $v_j(X_j) = b_1$, $v_j(X_i) = b_2$, $|E_{ij} \setminus (X_i \cup X_j)| = k$, and finally $\min_{e \in X_j}\{v_i(e)\} = c$. We can compute the values of L_{ij}^i via simple dynamic programming.

If t is a forget node with a child t' such that $B_{t'} = B_t \cup \{i\}$ for some $i \notin B_t$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 36. *For all $\sigma = (r, d, k) \in \Sigma_t$: $val_t((r, d, k)) = 1$ if and only if the following holds:*

There is $\sigma' = (r', d', k') \in \Sigma_{t'}$, for all $j \in B_t$ there are $(a_j^1, a_j^2, b_j^1, b_j^2, c_j, k_j) \in [0, W]^4 \times [0, W] \cup \{\infty\} \times [0, 2W+1]$ verifying one of the following three points:

1. - $d'(i) = Some$.
   - *for all $j \in B_t$, $a_j^1 = b_j^2 = 0$.*
   - *for all $j \in B_t$, if $d'(j) = Some$ then $k_j = |E_{ij}|$.*
2. *There is $j' \in B_t$, $c \in [0, W]$ such that:*
   - $r'(i) = None$.
   - $d'(i) \neq Some$.
   - $d'(j') \neq Some$.
   - $L_{ij'}^{j'}((b_{j'}^1, b_{j'}^2), (a_{j'}^1, a_{j'}^2), c, k_{j'}) = 1$.
   - *for all $j \in B_t \setminus \{j'\}$, $a_j^1 = b_j^2 = 0$.*
   - *for all $j \in B_t \setminus \{j'\}$, if $d'(j) = Some$ then $k_j = |E_{ij}|$.*
   - $d(j') = \max\{d'(j'), b_{j'}^2 - c\}$.
3. - $d'(i) \neq Some$.

and the following:

- $val_{t'}(\sigma') = 1$.
- $k = k' + \sum_{j \in B_t} k_j$
- *for all* $j \in B_t$, $L^i_{ij}((a^1_j, a^2_j), (b^1_j, b^2_j), c_j, k_j) = 1$
- *for all* $j \in B_t$,
  - *if* $d(j) \neq$ *Some:* $d'(j) \neq$ *Some,* $d(j) = \max\{d'(j), b^2_j\}$ *(except if this value has already been defined in case 2.), and* $a^2_j \leq r'(i) + \sum_{j \in B_t} a^1_j$, *and if* $c_j \neq \infty$, $r(j) = \min\{r'(j) + b^1_j, W\}$, *otherwise* $r(j) = r'(j)$.
  - *if* $d(j) =$ *Some and* $d'(j) \neq$ *Some: then* $r(j) \leq b^1_j$, $r'(j) =$ *None,* $r(j) = b^1_j$ *and* $a^2_j - c_j \leq r'(i) + \sum_{j \in B_t} a^1_j$.
  - *otherwise* $d(j) = d'(j) =$ *Some,* $c_j = \infty$, $r(j) = r'(j)$ *and* $r(j) \geq b^2_j$.
- *if* $d'(i) \neq$ *Some, then* $d'(i) \leq r'(i) + \sum_{j \in B_t} a^1_j$

While the details of this lemma seem complex, the intuition behind it is fairly straightforward: when we have to treat vertex i, we have to choose the orientation of the edges it shares with B_t.

In case **1.**, i is envied by one vertex of V_t. Therefore, it cannot receive anything from any vertex of B_t, and all the edges between i and B_t must be oriented towards B_t.

In case **2.**, we choose that i is envied by one vertex j of B_t. Therefore, it must not have received anything from any other vertex, we must orient the edges not shared with j towards B_t.

In case **3.**, i is not envied by any vertex of V_t, and we choose that it won't be envied by any vertex of B_t either. Therefore, we choose an orientation of the edges between i and B_t where i will be non strongly envious of some vertices of B_t, and verify that it received enough to be non-strongly envious of the vertices of V_t.

Finally, we check that the values of the two signatures and our chosen orientation correspond.

Proof. We note that $V_t = V'_t \cup \{i\}$, and $E_t = E_{t'} \cup \bigcup_{j \in B_t} E_{ij}$. Also, $V_t \cup B_t = V_{t'} \cup B_{t'}$.

In that case, a partial orientation $O \in A_t$ can be seen as a partial orientation $O' \in A_{t'}$, together with a partial orientation O'' of $(B'_t, \bigcup_{j \in B_t} E_{i,j})$. In the following, we show that given a partial orientation $O \in A_t$, its signature $\mathrm{sgn}_t(O)$ can be computed using only the following information: the signature of O' the restriction of O to $G_{t'}$, and the orientation of E_{ij} fo all $j \in V_t$ of O.

Let $O = \{X_j\}_{j \in V_t \cup B_t} \in A_t$.

For all $j \in B_t$, let $\alpha_j := E_{ij} \cap X_i$, $\beta_j := E_{ij} \cap X_j$, and $X'_j := X_j \setminus E_{ij}$: α_j is the set edges of E_{ij} received by i while β_j is the set of the ones received by j. X'_j is the set of edges received by j from $G_{t'}$.

Let $X'_i := X_i \setminus (\bigcup_{j \in B_t} E_{ij})$ and for all $\ell \in V_{t'}$, let $X'_\ell := X_\ell$. $O' := \{X'_j\}_{j \in B_{t'} \cup V_{t'}}$ is a partial orientation of $G_{t'}$, the restriction of O to $G_{t'}$, so $O' \in A_{t'}$. Let $\sigma = (r, d, k) = \mathrm{sgn}_t(O)$ and $\sigma' = (r', d', k') = \mathrm{sgn}_{t'}(O')$.

We note that $k = k' + \sum_{j \in B_t} |E_{ij} \setminus (\alpha_j \cup \beta_j)|$.

For all $j \in B_t$, let $a^j_1 := v_i(\alpha_j)$, $a^j_2 := v_i(\beta_j)$, $b^j_1 := v_j(\alpha_j)$, $b^j_2 := v_j(\beta_j)$, and $c_j = \min_{e \in \beta_j}\{v_i(e)\}$. By linearity of the value functions, we have: $v_i(X_i) = v_i(X'_i) + \sum_{j \in B_t} a^1_j$, and for all $j \in B_t$,

$v_j(X_j) = v_j(X'_j) + b_1^j$, $v_i(X_j) = v_i(X'_j) + a_2^j = a_2^j$, $v_j(X_i) = v_j(X'_i) + b_2^j = b_2^j$ and $k_j = |E_{ij} \setminus (\alpha_j \cup \beta_j)|$. By definition, $L_{ij}^i((a_j^1, a_j^2), (b_j^1, b_j^2), c_j, k_j) = 1$ and $k = k' + \sum_{j \in B_t} k_j$.

If $O \in A_t^{ef}$, then it follows immediately that $O' \in A_{t'}^{ef}$, and therefore, $val_{t'}(\sigma') = 1$. Furthermore, O makes i non-strongly envious of any other vertex. Finally, from Observation 34, either i is envied (but not strongly) by a unique vertex of V_t and therefore only receives edges shared with that vertex, or no vertex of $V_{t'}$ envies i, and either it receives edges from a unique vertex of B_t, or it doesn't. We will treat those three cases separately, which will result in the three cases of the lemma.

First, suppose that i is envied (but not strongly) by a vertex of V'_t in O. Then, it also holds in O'. Therefore, $d'(i) = Some$. We show that it implies point 1.

In that case, i does not receive any edges shared with vertices of B_t, so, for all $j \in B_t$, $a_j^1 = b_j^2 = 0$. Let $j \in B_t$, if $k_j < |E_{ij}|$, j receives at least one edge from i in O, and therefore, cannot be envied by a vertex of $V_{t'}$, so $d'(j) \neq Some$.

Second, suppose that i only receives edges shared with a vertex $j' \in B_t$. Then, it does not receive any edges shared with vertices of $V_{t'}$. Therefore, $r'(i) = None$ and $d'(i) \neq Some$. We show that it corresponds to case 2. Let $c = \min_{e \in \alpha_{j'}}\{v_{j'}(e)\}$. By definition, $L_{ij'}^{j'}((b_{j'}^1, b_{j'}^2)(a_{j'}^1, a_{j'}^2), c, k_{j'}) = 1$.

For all $j \in B_t \setminus \{j'\}$, i does not receive any edge shared with j, and therefore $a_j^1 = b_j^2 = 0$. Also, if $k_j < |E_{ij}|$, j receives at least one edge from i in O, and therefore, cannot be envied by a vertex of $V_{t'}$, so $d'(j) \neq Some$. i did not receive any edge from any vertex other than j', and therefore $d(j') = \max\{d'(j'), b_{j'}^2 - c\}$. Finally, j' receives at least one edge from i, and therefore $d'(j) \neq Some$.

Third, suppose that we are not in the two previous cases. Then, $d'(i) \neq Some$. It implies case 3 trivially.

We prove the remaining points: let $j \in B_t$:

If $d(j) \neq Some$: then $d'(j) \neq Some$ by definition, and if i received edges from vertices other than j in O, then $d(j) = \max\{d'(j), b_j^2\}$. Furthermore, i is non-envious of j, so $a_j^2 \leq r'(i) + \sum_{j \in B_t} a_j^1$; finally, if $c_j \neq \infty$, then j receives at least one edge from E_{ij} and therefore $r(j) = \min\{r'(j) + b_j^1, W\}$; otherwise, $r(j) = r'(j)$.

If $d(j) = Some$ and $d'(j) \neq Some$: then i is envious of j but not strongly envious; therefore, j does not receive any edge shared with any vertex of $V_{t'}$, so $r'(j) = None$ and $r(j) = b_j^1$. Furthermore, j is non-envious of i, so $r(j) \leq b_j^1$. Finally, i is non strongly envious of j, so $a_j^2 - c_j \leq r'(i) + \sum_{j \in B_t} a_j^1$.

Otherwise, if $d(j) = d'(j) = Some$, then j is envied (but not strongly) by a vertex of $V_{t'}$. Therefore, j does not receive any edge of E_{ij}, so $c_j = \infty$ and $r(j) = r'(j)$. Also, j does not envy i, and therefore $r(j) \geq b_j^2$.

Finally, i is not strongly envious of the vertices of $V_{t'}$, and therefore, if $d'(i) \neq Some$, $d'(i) \leq r'(i) + \sum_{j \in B_t} a_j^1$.

□

Join node. If t is a join node with two children t_1, t_2 such that $B_t = B_{t_1} = B_{t_2}$, the following lemma allows us to compute the values of val_t over Σ_t dynamically:

Lemma 37. *For all $\sigma = (r, d, k) \in \Sigma_t$: $val_t((r, d, k)) = 1$ if and only if there are $\sigma_1 = (r_1, d_1, k_1) \in \Sigma_{t_1}$ and $\sigma_2 = (r_2, d_2, k_2) \in \Sigma_{t_2}$ such that:*

- $val_{t_1}(\sigma_1) = 1$ *and* $val_{t_2}(\sigma_2) = 1$
- $k = k_1 + k_2$
- *for all* $j \in B_t$,
  - *if* $d_1(j) = Some$ *(respectively* $d_2(j) = Some$*) then* $r_2(j) = None$ *(respectively* $r_1(j) = None$*),* $d(j) = Some$*,* $d_2(j) \leq r_1(j)$ *(respectively* $d_1(j) \leq r_2(j)$*) and* $r(j) = r_1(j)$ *(respectively* $r(j) = r_2(j)$*).*
  - *otherwise, if* $r_1(j) = r_2(j) = None$ *then* $r(j) = None$ *and* $d(j) = \max\{d_1(j), d_2(j)\}$
  - *otherwise,* $r(j) = \min\{r_1(j) + r_2(j), W\}$ *and* $d(j) = \max\{d_1(j), d_2(j)\}$

Intuitively, a valid partial orientation of G_t is a combination of two valid partial orientations of G_{t_1} and G_{t_2}; therefore, we can combine two possible valid partial orientations of G_{t_1} and G_{t_2} and see if they are compatible, that is, if some vertex of B_t is envied by one vertex of V_{t_1}, it must not receive any edge from G_{t_2}, and conversely. In that case, we can combine the two orientations into an orientation of G_t.

Proof. Note that $V_t = V_{t_1} \cup V_{t_2}$, with $V_{t_1} \cap V_{t_2} = \varnothing$. Similarly, $E_t = E_{t_1} \cup E_{t_2}$, with $E_{t_1} \cap E_{t_2} = \varnothing$.

Let $O = \{X_j\}_{j \in \{B_t \cup V_t\}} \in A_t$; for all $j \in B_{t_1} \cup V_{t_1}$, let $X_j^1 := X_j \cap E_{t_1}$ and for all $j \in B_{t_2} \cup V_{t_2}$, let $X_j^2 := X_j \cap E_{t_2}$.

Let $O_1 := \{X_j\}_{j \in \{B_{t_1} \cup V_{t_1}\}}$ and $O_2 := \{X_j\}_{j \in \{B_{t_2} \cup V_{t_2}\}}$. They are the restrictions of O to G_{t_1} and G_{t_2} respectively: $O_1 \in A_{t_1}$ and $O_2 \in A_{t_2}$.

Let $\sigma = (r, d, k) = \mathrm{sgn}_t(O)$, $\sigma_1 = (r_1, d_1, k_1) = \mathrm{sgn}_{t_1}(O_1)$ and $\sigma_2 = (r_2, d_2, k_2) = \mathrm{sgn}_{t_2}(O_2)$. By definition, $k = k_1 + k_2$, $val_1(\sigma_1) = 1$ and $val_2(\sigma_2) = 1$.

Let $j \in B_t$. If $d_1(j) = Some$, some vertex of V_{t_1} envies j, so $d(j) = Some$. Furthermore, in that case, j does not receive any edge shared with any other vertex, and so $r_2(j) = None$, and $r(j) = r_1(j)$. Finally, j does not envy any vertex from V_{t_2}, so $d_2(j) \leq r_1(j)$. The same holds symmetrically if $d_2(j) = Some$.

Otherwise, if $r_1(j) = r_2(j) = None$, then j does not receive any edge shared with vertices of V_t, and therefore $r(j) = None$. Furthermore, no vertex of V_t is envious of j, so $d(j) = \max\{d_1(j), d_2(j)\}$.

Finally, otherwise, j receives at least one edge shared with vertices of V_t, so $r(j) = \min\{r_1(j) + r_2(j), W\}$, and no vertex of V_t is envious of j, so $d(j) = \max\{d_1(j), d_2(j)\}$. □

Concerning the complexity, supposing the values of val_{t_1} and val_{t_2} have already been computed, there are $m \times W^{\mathcal{O}(\mathrm{tw})}$ signatures of t_1 and t_2 to consider. Therefore, this step can be done in time $mW^{\mathcal{O}(tw)} \times mW^{\mathcal{O}(tw)} = m^2 W^{\mathcal{O}(tw)}$.

This concludes the description and correctness of our algorithm. We now argue about its complexity: as the nice tree decomposition can be supposed to be of polynomial size and each step can be done in time at most $m^2 W^{\mathcal{O}(\mathrm{tw})}$, the algorithm has complexity $W^{\mathcal{O}(\mathrm{tw})}(nm)^{\mathcal{O}(1)}$. □